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GREEN BANKING AWARENESS AND FINTECH ADOPTION TOWARDS PROFITABLE AND SUSTAINABLE PRACTICES

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ABSTRACT

The research aims to investigate the effect of green banking awareness (GBA) and green FinTech adoption (GFA) on sustainable financial performance in the banking industry. The specific aim of the research is to investigate the effect of green banking awareness on green FinTech adoption, and the effect of green FinTech adoption on ESG investment decisions, returns/profitability, and sustainable banking practices. The research will also investigate the mediating effect of green FinTech adoption in the relationship between green banking awareness and sustainable financial performance. The study's research design will be descriptive and causal. The descriptive part of the research will examine respondents' perceptions of green banking awareness and green FinTech adoption. The study will use summary statistics to determine respondents' perceptions. The causal part of the research will examine the direct and indirect relationships among the variables. The study targets banking professionals, and data will be collected via a structured questionnaire using a five-point Likert scale. The instrument was developed based on a literature review and expert opinions, thereby establishing its validity and reliability. The data analysis was conducted using SPSS, which enabled descriptive statistics, correlation analysis, and multiple regression. The analysis of the mediating effects of green FinTech adoption was conducted using Hayes's PROCESS Macro procedure. The results of the regression analysis indicate that green banking awareness has a strong, positive, and significant influence on ESG investment decisions, returns/profitability, and sustainable banking practices. On the other hand, green FinTech adoption has a positive and significant influence on the aforementioned variables. It can be concluded that green banking awareness is an important factor that needs to be strengthened to achieve sustainable financial performance. In addition, green FinTech adoption needs to be enhanced to achieve sustainable financial performance. Policymakers need to enhance green innovation and sustainability awareness to achieve sustainable financial performance.

Keywords: *Green Banking awareness, Green Fintech Adoption, Environmental, Social and Governance, Sustainable Practices*

CHAPTER I

INTRODUCTION

1.1 Background of the Study

The world is witnessing a revolution in its financial system driven by two major factors: the rapid pace of technological change and the need to address environmental sustainability challenges. In the last ten years or so, the world has witnessed a revolution in financial technology, transforming the way financial services are delivered to customers through increased efficiency, accessibility, transparency, and cost-effectiveness. The use of digital platforms, blockchain technology, mobile banking, artificial intelligence, and data analytics has completely changed the way financial institutions operate (Wang et al., 2021).

Meanwhile, environmental issues such as climate change, carbon footprints, biodiversity loss, and resource depletion have heightened global interest in responsible financial practices. Financial institutions are now seen as important drivers in achieving the Sustainable Development Goals (SDGs) by mobilizing funds. To achieve this, green finance was conceptualized as a strategic approach to integrate environmental concerns into financial decisions. Green finance involves investing in environmentally friendly projects, such as renewable energy, sustainable agriculture, clean transportation, and climate adaptation (Zhang et al., 2022). In this process, green banking is a key concept that integrates environmental concerns into traditional banking practices (Gupta, 2025).

On the other hand, green banking refers to environmentally friendly banking practices aimed at reducing carbon footprints. Green banking practices include paperless banking transactions, green loans, renewable energy finance, carbon footprint measurement, and ESG-based risk assessment. Through these practices, banks can influence corporate behavior and promote sustainable production and consumption patterns (Yan et al., 2022). As financial intermediaries, banks have the power to mold the path of economic development through their investment choices.

The integration of FinTech and green finance has led to the emergence of a new phenomenon: green FinTech, a combination of digital innovation and sustainability. In this case, green FinTech uses digital technology to increase the transparency, traceability, and efficiency of green financial flows. For instance, blockchain technology is used to increase accountability in the issuance of green bonds, artificial intelligence is used to increase environmental risk assessment, and digital platforms are used to increase access to green investment opportunities (Mirza et al., 2023). The empirical literature has shown that the use of FinTech has a positive impact on green lending, ESG investment, and financial and environmental performance (Yan et al., 2022; Mirza et al., 2023). Therefore, green FinTech is an infrastructure that enables the rapid financial transition to sustainability.

However, technological innovations by themselves do not automatically imply sustainability. The sustainability of green FinTech depends to a large extent on stakeholders' awareness and understanding. Green banking awareness is the extent to which customers, investors, managers, and policymakers understand the environmental impacts of financial activities and the benefits of sustainable banking. Awareness shapes attitudes, trust, and behavioral intentions towards the adoption of green financial technologies (Sharma et al., 2025). Empirical research has shown that stakeholders with greater environmental knowledge are more likely to support the adoption of green financial innovations (Tanchangya et al., 2025). Therefore, awareness is the cognitive and behavioral basis on which sustainability is built.

In emerging economies, the significance of green banking and FinTech adoption is even greater. In developing countries, access to green finance is usually limited, transaction costs are higher, and infrastructure is limited. Digital financial technologies provide solutions to these problems at a relatively low cost. This would not only improve the overall performance of these economies but also benefit the environment (Nguyen et al., 2024).

In Nepal, the significance of green banking and the integration of FinTech is particularly pronounced. The country is more vulnerable to the impacts of climate change, including glacier melting, floods, landslides, and unpredictable weather. Sustainable finance is critical in supporting the development of renewable energy,

climate resilience, and environmentally friendly economic growth. The Nepal Rastra Bank (NRB, 2022) has issued guidelines to promote green lending, energy-saving practices, and sustainable risk management practices among financial institutions. At the same time, mobile banking and QR code transactions are gaining traction in the country.

Nevertheless, the integration of green finance principles into digital banking practices remains limited. Public awareness of green banking initiatives is relatively low, and many financial institutions have yet to fully integrate ESG considerations into their digital platforms. While studies suggest that FinTech adoption positively influences sustainability performance (Joshi & Karmacharya, 2024), empirical research examining the interconnected role of green banking awareness and green FinTech adoption in Nepal remains scarce.

Moreover, financial institutions have concerns about whether sustainable initiatives are profitable. Although studies show that digitalization and innovation can improve operational efficiency and profitability (Mirza et al., 2023; Nguyen et al., 2026), financial institutions in developing economies have concerns about whether green investments are profitable. It is important to understand the relationship between green FinTech adoption and investment decisions based on profitability and sustainable banking practices.

In light of the above, an analysis of the green banking awareness factor with respect to green FinTech adoption and sustainability results will not only be timely but also relevant. This study aims to contribute to the existing literature on sustainable finance by conducting an empirical analysis of the Nepalese banking industry. By analyzing the relationships among awareness, digital technology, ESG, profitability, and sustainability, this study aims to provide insights to inform policy and institution-building for sustainability.

1.2 Problem Statement

The Nepalese banking industry is witnessing a digital revolution with rapid changes in its regulatory environment, technological advancements, and increasing environmental concerns. Nepal Rastra Bank (2022) has encouraged banks to promote

sustainable practices through "paperless banking," "energy-efficient banking," and "environmentally conscious lending." However, public awareness of green banking practices is low. People are unaware of green financial products, ESG investment principles, and the environmental impact of digital financial services (Bansal et al., 2023; Bouteraa et al., 2023).

While FinTech has shown promise in supporting green finance, ESG investments, and sustainable banking performance (Mirza et al., 2023; Yan et al., 2022), low digital literacy levels, infrastructure challenges, and general hesitancy towards change also constrain its adoption in Nepal. Accordingly, the potential of FinTech in supporting green banking performance has yet to be fully harnessed.

In addition, there is little research done in Nepal that investigates the role of awareness of green banking on the adoption of green FinTech, as well as the effect of the latter on ESG investment decisions, profitability, and sustainable banking practices. Existing studies have mainly focused on the adoption of FinTech or sustainability individually, without considering their joint effects. This issue has been recognized by Joshi & Karmacharya (2024). The role of green FinTech adoption as a mediator between awareness of green banking and sustainability outcomes, especially in developing countries, remains underexplored (Galeone et al., 2024; Mirza et al., 2023).

Moreover, banks perceive green initiatives as cost-increasing rather than value-creating, creating uncertainty about financial returns. In the absence of empirical evidence on financial returns and sustainability benefits, financial institutions are unlikely to invest more in green FinTech.

Thus, there is a research gap regarding the integrated role of green banking awareness and green FinTech adoption in relation to ESG investment behaviors, financial returns, and sustainable banking practices in Nepal. Besides, the research addresses the following issues:

- What is the impact of green banking awareness on ESG-based investment Decisions?

- What is the impact of green banking awareness on perceived returns and profitability?
- What is the influence of green banking awareness on sustainable banking practices?
- Does green fintech adoption mediate the relationship between green banking awareness and sustainable banking?

1.3 Objectives of the Study

- To examine the impact of green banking awareness on the adoption of green FinTech solutions.
- To analyze the effect of green FinTech adoption on ESG-based investment decisions.
- To evaluate the influence of green FinTech adoption on perceived returns and profitability.
- To examine the role of green FinTech adoption in promoting sustainable banking practices.
- To assess the mediating role of green Fintech adoption on Environment, Social and Governance, Perceived Returns/profitability, and Sustainable banking.

1.4 Rationale of the Study

The financial industry is undergoing tremendous change in two forms: rapid digitalization and the need to adopt sustainable practices. In one way, technology in finance has revolutionized how banks provide services to customers. In another way, environmental issues such as global warming, resource depletion, and pollution pose a threat to sustainable development. Therefore, there is a need to understand the connection between digitalization and sustainable practices in the banking industry.

Although the scope of digital banking services has been increasing rapidly, their potential contribution to sustainability has not been adequately explored, especially in developing economies like Nepal. Many financial organizations are investing in digital banking, but the integration of environmental principles into financial technology has not been sufficiently explored. Moreover, awareness about green banking practices is also quite low. Without sufficient awareness, neither customers

nor financial managers can understand the potential benefits of green financial technology. Therefore, it becomes crucial to understand the significance of green banking awareness in motivating the adoption of green financial technology.

Another important reason for conducting this study is the uncertainty about the profitability of sustainable initiatives. While sustainable banking can be considered socially responsible, many institutions are uncertain whether sustainable initiatives are actually profitable. Banks are unlikely to invest in green technology because they are unsure whether it is profitable enough. By investigating the relationship between green FinTech adoption and returns and profitability, this study aims to provide evidence on whether sustainability and profitability are mutually exclusive concepts.

This study is especially important in Nepal, a country facing many environmental challenges, including climate-related disasters and energy needs. At the same time, it is a country where financial technologies, including mobile banking and online transactions, are increasingly used. The financial sector in Nepal is at a critical juncture, at which financial technology can either continue on its current trajectory as a simple technological extension or become a transformative force towards a more sustainable financial sector. The role of awareness in adoption, and the role of adoption in investment decisions based on ESG criteria and sustainable banking, can help in a more positive transformation.

Moreover, there is a research gap in integrated research on the banking industry in Nepal. The earlier research has focused more on digital adoption or sustainability practices separately. Few studies have examined the interconnectedness among green awareness, digital adoption, profitability, ESG, and sustainable banking practices. This research aims to fulfill this research gap by conducting a comprehensive analysis of the interconnectedness of the aforementioned variables.

Practically speaking, the results of this study can help policymakers, regulators, and bank managers develop strategies. If green banking awareness is proven to play a vital role in green FinTech adoption, then greater emphasis can be placed on raising awareness. Similarly, if green FinTech adoption can boost bank profitability and sustainable banking practices, banks can be encouraged to invest in sustainable

technologies. Therefore, this study provides practical benefits that can contribute to economic growth while maintaining environmental sustainability.

In conclusion, it can be said that the current study is timely and necessary in addressing the urgent call for sustainable financial systems, contributing to the understanding of the role of digital innovation in environmental responsibility, and providing empirical findings pertinent to the changing Nepalese banking landscape in relation to awareness, adoption, profitability, and sustainability. The research aims to support a more resilient, environmentally responsible financial future.

1.5 Hypothesis of the study

In order to examine the relationship between green banking awareness, green FinTech adoption, and sustainable financial outcomes, the following hypotheses have been developed:

H1: Green banking awareness has a significant impact on investment decisions based on ESG.

H2: Green banking awareness has a significant impact on perceived returns and profitability.

H3: Green banking awareness has a significant impact on sustainable banking practices.

H4: Green fintech adoption has a significant impact on investment decisions based on ESG.

H5: Green fintech adoption has a significant impact on perceived returns and profitability.

H6: Green fintech adoption has a significant impact on sustainable banking practices.

H7: Green fintech adoption mediates the relationship between green banking awareness and investment decisions based on ESG.

H8: Green fintech adoption mediates the relationship between green banking awareness and perceived returns and profitability.

H9: Green fintech adoption mediates the relationship between green banking awareness and sustainable banking practices.

1.6 Limitations of the study

While this research provides valuable insights into green banking and FinTech in Nepal, several limitations should be kept in mind when reviewing the findings.

- **Cross-sectional design:** The study is conducted at a single point in time; that is, it provides a snapshot of perceptions and relationships at a single point in time. While this design helps understand short-term changes, it is less effective for understanding long-term changes. For example, we cannot determine how awareness of green banking leads to long-term FinTech adoption.
- **Use of self-reported data:** The results are based on the data collected through questionnaires from the participants. The participants may have given social desirability bias in their answers, especially on sensitive issues such as sustainability and environmental responsibility. This may cause minor biases in the results.
- **Limited geographical scope:** The focus is on the banking sector in Nepal, which is appropriate in a developing economy context, but it may not be universally applicable. The results may differ in countries with stronger regulations, better technology, and greater awareness of environmental issues.
- **Limited number of variables:** The study focuses specifically on factors like green banking awareness, adoption of green FinTech, ESG-based investment decisions, profitability, and sustainability. It does not go into great detail about other factors such as regulatory pressure, digital readiness, organizational culture, or customer trust, which may be significant.
- **Moderation model focus:** In this regard, by considering green FinTech adoption as a mere moderator, it can be stated that the analysis of interaction effects is effectively carried out. Nevertheless, other methods, such as complete mediation or structural equation modeling, could be adopted to examine nuances of the relationship between these factors.
- **Rapid technological and policy changes:** Nepal's financial environment is constantly changing, with digital innovations and regulatory updates frequently coming into the scene. While this study captures the environment at the time of data collection, recent innovations may significantly influence the role of green FinTech in sustainability.

1.7 Organization of the study

This thesis is structured into five chapters to ensure a logical and coherent presentation of the research.

The first chapter introduces the study, including background information and context on green banking awareness and green FinTech adoption in the banking sector. The research problem, objectives, questions, and hypotheses are presented, along with the study's significance, scope, and limitations. The concepts and variables are clearly defined, creating a robust foundation for analysis.

The second chapter involves an extensive literature review on green banking, financial technology, ESG investment decisions, profitability, and sustainable banking practices. This chapter examines existing theories and empirical studies on the subject. The literature gap is also addressed, creating a conceptual framework for analysis.

The third chapter of the thesis discusses the research methodology adopted for the study. This chapter discusses the research design adopted for the study. The sampling method adopted for the study is also discussed in this chapter. The methods used to collect data for the study are also discussed in this chapter. Furthermore, the statistical methods used to test the hypotheses are also discussed in this chapter.

The fourth chapter of the thesis analyzes the collected data and presents the study's findings. This chapter presents the study's findings. The study's findings are presented in relation to its objectives.

The fifth chapter of the thesis presents the study's significant findings. This chapter presents the study's findings. The conclusions of the study are also presented in this chapter. Furthermore, the study's implications are presented in this chapter.

CHAPTER II

LITERATURE REVIEW

The literature review of this study will be a comprehensive and systematic analysis of the concepts, terminology, studies, theories, and gaps related to awareness of green banking, the adoption of green FinTech, and their contributions to profitable yet sustainable practices. This study will be a synthesis of various scholarly articles and journals. This study will analyze how responsible banking practices and FinTech services contribute to sustainability while promoting economic growth. This study will include scholarly articles that analyze the relationship between green financial practices and business profitability, demonstrating that sustainability practices can go hand in hand with economic growth.

Despite the ever-increasing number of publications in the field of study, the review noted research gaps, particularly in the integrated study of green banking awareness and green FinTech adoption as twin drivers of profitable and sustainable business practices, especially in developing country contexts. There is limited research examining the relationships among awareness, technology adoption, and their simultaneous impact on financial profitability and environmental sustainability. This study aims to contribute to the existing body of knowledge by addressing these research gaps.

2.1 Conceptual Review

The conceptual review of this study primarily focuses on the essential concepts of green finance, corporate social responsibility (CSR), green investment, and sustainable business performance, which are integral to explaining environment-conscious business and financial practices. Green finance refers to a set of financial instruments and strategies that incorporate environmental considerations into economic activities to ensure sustainable development while maintaining financial viability. Corporate social responsibility refers to an organization's moral obligation to conduct its operations in a socially responsible manner, be accountable to society, and be environmentally conscious. Green investment refers to an organization's strategic effort to allocate resources to environmental protection initiatives. Sustainable business performance refers to an organization's ability to achieve economic growth while minimizing its environmental impact and promoting

environmental sustainability. These concepts provide an essential foundation for understanding the relationships between environmentally conscious financial practices and business performance, and for assessing the potential impact of an organization's financial decisions on sustainable growth.

2.1.1 Green Banking Awareness

Green finance has become a key term in modern financial and environmental debates, as ecological sustainability is integrated into financial decision-making and investment processes. In essence, green finance is the financial activities, instruments, and services that enable the flow of capital to environmentally positive initiatives while promoting economic growth. Current research has described green finance as a mechanism to balance ecological sustainability with economic growth, incorporating measures to mitigate climate change. Current research has shown that green finance comprises a wide range of instruments, including green bonds, sustainability-linked loans, and products for mitigating climate risk, that enable the flow of resources to environmentally sustainable initiatives (Zioło et al., 2025).

Furthermore, extensive bibliometric research shows that the development of environmental concerns, regulatory reforms, and the growing significance of sustainable investment frameworks have influenced the conceptual development of green finance. Experts highlight that green finance is not limited to investing in renewable energy sources and low-carbon infrastructure; it also involves incorporating environmental, social, and governance criteria into mainstream financial practices to transform the dynamics of financial institutions. Furthermore, the interface of green finance with financial technologies has also created a new research paradigm, 'climate FinTech', that focuses on applying digital technologies to enhance the assessment of environmental impacts by increasing transparency in sustainable financial products (Zioło et al., 2025).

However, despite the wide conceptual scope and applicability of green finance, there are challenges to implementing it. Firstly, differences in the definitions and taxonomies of green finance across jurisdictions pose challenges to standardizing it. Additionally, information asymmetry and differences in regulatory support also affect

the rate at which green finance can mobilize financial resources to sustainable initiatives (Zioło et al., 2025). This problem is more pronounced in emerging markets, which lack advanced frameworks to identify green finance initiatives despite the growing demand.

Nonetheless, green finance is a critical component of global efforts to achieve sustainability. By mobilizing private and public capital into environmentally friendly sectors of the economy, green finance is critical to environmental protection, mitigating the impacts of climate change, and fostering sustainable economic growth (Cui et al., 2020). The continued development of standardized sustainable finance practices is critical to improving the efficiency of green finance in fostering environmental sustainability. Green finance is a critical component of global financial markets and environmental policy discussions, as it fosters sustainable development and mitigates the impacts of climate change. The concept of green finance has evolved, leading to diverse definitions that reflect its multidimensional and dynamic nature in the global economy (Lee, 2020). In essence, green finance is a broad term that describes financial activities that incorporate environmental factors into investment and credit decisions to foster environmentally sustainable growth.

At its most basic level, green finance can be understood as financial investments, instruments, and services that are committed to protecting the environment and promoting sustainable development. According to Guild (2020), "Green finance is the allocation of economic resources to projects, initiatives, and policies that contribute to a sustainable economy. Green finance goes beyond climate finance to include biodiversity conservation, energy efficiency, and other activities that are beneficial to the environment." This broad description underscores the potential impact of green finance in different sectors of the economy.

Expanding on the definition, Bloomberg New Energy Finance clarifies that green finance also includes expenditures on project development, such as conducting feasibility studies and acquiring land. Green finance also recognizes the complexity involved in project finance in promoting green initiatives (Cui et al., 2020).

This definition further emphasizes the importance of having a comprehensive financing system in supporting green initiatives. From an institutional viewpoint,

PricewaterhouseCoopers emphasizes the role of financial institutions in integrating environmental criteria into their lending decisions, thereby aligning economic incentives with ecological sustainability (Volz et al., 2015).

Moreover, Volz et al. (2015) highlighted the importance of environmental screening and risk assessment in defining what is meant by 'green finance' in terms of investment portfolios. Such an approach ensures not just that financial investments are made in accordance with sustainability criteria, but also that they have a positive impact on the environment, such as reduced carbon emissions. Similarly, Lindenberg (2014) has adopted a broader conceptual approach, which includes reference to public policies and regulations to support 'green investments' and combat environmental degradation, and the interrelationship between markets, policy, and financial institutions.

More recent research by Guild (2020) distinguishes between green financing, which is geared towards achieving specific environmental and climate-related goals, and the 'greening of finance,' which is the incorporation of sustainability principles into conventional financial practice. The distinction results from continuous efforts to mainstream environmental considerations across the broader financial system.

Despite its conceptual maturity and growing global applicability, green finance is also subject to several implementation challenges. Information asymmetry is one challenge to green finance. Information asymmetry acts as a barrier to investors' ability to assess the environmental risks and returns of green investments properly. Differences in the definitions and classification systems for green financial product categories also pose a challenge for green finance. Differences in political support are another factor that influences the impact of green finance initiatives. Weak policy frameworks and low incentives are challenges to green finance in some regions (Lee, 2020).

In developing and emerging countries, the availability of capital may not be as significant an issue as the challenge of developing financially viable, environmentally friendly projects. Guild (2020) highlights the importance of developing specific financial instruments and regulatory frameworks to support green finance, especially in developing countries, where sustainability issues are closely linked to overall socioeconomic development. Nevertheless, it is critical to understand that green

finance plays an essential role in achieving global sustainability goals. By mobilizing capital into renewable energy, green finance promotes economic growth while protecting the environment and improving climate resilience.

Moving forward, the future of green finance will depend on removing current hurdles through increased transparency, the use of standardized environmental impact measurements, and the strengthening of existing regulatory frameworks that foster green investments. This will require increased collaboration among governments, financial bodies, and the private sector to scale up green finance projects and align them with global sustainability agendas, including the United Nations Sustainable Development Goals (SDGs). In conclusion, green finance has emerged as a vital tool for promoting sustainability, thereby strengthening a global financial system that is both strong and sustainable (Cui et al., 2020).

2.1.2 FinTech Adoption

In recent years, the interface between FinTech and Environmental, Social, and Governance (ESG) investing has emerged as a prominent area of research, reflecting the dynamic nature of sustainable finance in digital contexts (Khan et al., 2025; Chen et al., 2025). Green FinTech platforms contribute to better ESG investment decisions by providing digital channels to process large data sets, enabling more robust analysis of ESG factors and more accessible sustainable investment options. For instance, investment portfolio personalization tools facilitated by FinTech technologies are found to significantly impact investors' intentions to use FinTech platforms for ESG investing, with perceived usefulness and ease of use identified as important factors influencing behavioral intention (Chen et al., 2025). In addition, studies provide evidence that FinTech adoption is positively associated with corporate ESG performance, suggesting that FinTech can alleviate financing constraints to improve ESG performance. Green FinTech platforms are also important for promoting ESG investment information to a wider range of global investors, enabling better access and more informed, ethical investment decisions. Blockchain technology, for instance, has been noted as helping build transparency and address risks such as misinformation and greenwashing in ESG investments, thereby fostering trust and accountability in the practice of sustainable finance (Khan et al., 2025). Green

FinTech, therefore, is seen not only as a supportive technology for ESG investment decisions but also as a core driver that helps propel them toward greater transparency, inclusiveness, and performance in line with sustainability goals, laying the groundwork for participants' confidence in these markets.

2.1.3 Profitability

Technological innovations in financial services are improving efficiency and profitability, including the adoption of green finance principles. According to Mirza et al. (2023), FinTech helps accelerate the adoption of green finance by enhancing risk-adjusted returns, reducing costs, and expanding the range of financial products. Another study by Lee (2020) found that banks that experience greater financial innovation are more likely to achieve higher profits, assets, and loans. Yang et al. (2021) emphasized that there is increased adoption of green finance principles in investment and financial decisions, a move toward sustainable development. Furthermore, the adoption of FinTech helps to promote access to sustainable products, address information asymmetry, and achieve quality economic growth (Nenavath & Mishra, 2023). All these studies indicate that adopting green finance technology benefits profitability and economic development.

The use of green financial technologies, or FinTech, has been shown to improve the profitability of financial institutions. This is because it enhances operational efficiency, lowers costs, and increases access to sustainable investment products. The use of technology to promote green finance helps financial institutions, such as banks and investment companies, achieve greater profitability while advancing sustainability. This can be achieved by optimizing risk-adjusted returns, diversifying product offerings, and increasing profitability while promoting sustainability. This concept has been supported by various studies, including those by Mirza et al. (2023) and Lee & Shin (2018). Studies have shown that financial institutions that have adopted green FinTech experience have achieved greater financial performance. This is because it helps to facilitate ESG-compliant investments, reduce information asymmetry, and increase capital inflows by attracting environmentally conscious investors. This helps increase the company's revenue flows. The use of FinTech tools such as blockchain and AI enhances transparency, thereby reducing the risks

associated with greenwashing. This helps to promote profitable investment decisions. (Khan et al., 2025).

2.1.4 Sustainable practices

With the increased global focus on sustainability, there has been a need for green finance and sustainable banking, further emphasizing the importance of Green Finance Adoption (GFA) as a means of promoting long-term sustainability (Guang-Wen & Siddik, 2022). In this context, recent studies have shown that integrating digital finance with sustainability has tremendous potential, as sustainable FinTech innovations can deliver more efficient financial services (Vergara & Agudo, 2021). In addition, FinTech innovations can positively impact socio-economic growth, including poverty reduction, thereby further enhancing sustainability (Vasishta & Singla, 2024). In line with global sustainability goals, including the Paris Climate Change Agreement and the United Nations Sustainable Development Goals (SDGs), policymakers in developing countries are now recognizing the importance of FinTech innovations, such as green finance, for promoting sustainability (Nassiry, 2018). In this context, banks have a major responsibility to adopt emerging technologies, including blockchain, online banking, and green banking, to promote sustainable financial practices (Gupta, 2025; Kalaiarasi & Kirubahari, 2023). They also help to finance environmentally friendly initiatives such as renewable energy, energy efficiency measures, clean technologies, and the development of green industries (Bei & Wang, 2023). The integration of green FinTech adoption by financial organizations and government policies facilitates the development of sustainable banking practices. This ensures that financial practices align with government regulations while promoting environmental sustainability.

The adoption and integration of sustainability in business practices have seen tremendous growth in recent times. This is evident in the significant expansion of sustainable investing, which accounted for \$35.3 trillion worldwide in 2020 (Cui et al., 2020). This shows a significant shift in investment practices, with environmental, social, and governance factors now considered alongside traditional financial factors in investment decisions. Sustainability in the corporate world is an integral component of business strategy that connects economic performance with social performance.

Sustainability challenges the traditional view that a corporation's only aim is to maximize profits for shareholders (Friedman, 2007), rather than to generate value for all stakeholders.

This transformation is heavily dependent on Corporate Social Responsibility (CSR) and Socially Responsible Investing (SRI). CSR involves businesses actively contributing to social and environmental causes, not just for financial gain (Kakabadse & Morsing, 2006). This is due to the increasing need for accountability and ethical business practices. Similarly, SRI incorporates ESG factors into investment decisions, enabling individuals to support sustainable practices while earning financial returns (Eurosif). This combination of ethical values and financial gains contributes to responsible investment practices (Lee, 2020).

2.2 Theoretical Underpinnings

2.2.1 Theory of green growth

The theory of green growth began to gain traction after the Rio+20 Conference of 2012, during which the outcome report, "The World We Want," emphasized the need to promote a green economy while supporting sustainable economic growth. Since then, the concept of green growth has become an integral part of strategies to promote environmental sustainability adopted by governments and multinational companies at both national and international levels. The theory of green growth, at its core, holds that it is possible to promote economic growth without increasing carbon emissions or resource consumption, thereby fostering sustainable growth. This theory of promoting economic growth without negative environmental consequences is critical for addressing climate change, underscoring the role of human capital as a catalyst for economic growth. This argument is further supported by studies on economic growth

conducted by experts, which emphasize that it is possible to promote economic growth while protecting the environment.

Investment is a key component of the green growth model. This is particularly true in the face of growing levels of consumption and production. While investments promote economic development in the country, they also contribute to higher CO₂ emissions when sustainability is ignored. To overcome this challenge, policymakers in Asian countries have increasingly focused on large-scale green investments. Estimates show that investments of about US\$1.7 trillion annually, totaling about US\$26 trillion between 2016 and 2030, are required to reduce poverty, promote economic development, and mitigate climate change. Investments in the green economy play a crucial role in transforming the energy sector by reducing dependence on non-renewable resources, such as coal, and shifting to renewable sources, such as wind, solar, and biofuels. In this regard, green finance is essential for channeling private capital into low-carbon, environmentally sustainable initiatives (Mo et al., 2023).

Nevertheless, the literature shows that carbon pricing alone is insufficient; green finance investment is required to be effective in addressing emissions and creating efficient energy systems (Mo et al., 2023). Therefore, the theory of green growth has been developed to incorporate green financial practices with supportive policies to achieve a low-carbon economy.

2.2.2 Theory of Green Finance

The theory of green finance helps close the gap between the financial sector and environmental protection efforts to achieve economic growth by preserving natural resources. It provides a framework for investing in the environment to achieve the goals of the green economy. This theory helps to promote investments that contribute to human development, environmental risks, and quality. For the financial sector to promote the coordinated development of resources, the environment, the economy, and society, it must follow the principles of environmental protection and sustainable development (Ye & Dela, 2023). Green finance has the potential to accelerate the

development of a low-carbon economy by encouraging companies to practice sustainable development to mitigate the impacts of climate change.

Moreover, green finance offers important economic co-benefits. The Global Commission on the Economy and Climate highlights the importance of green finance in mobilizing substantial private capital to support climate-friendly investments and activities (Schmidt et al., 2017). The incorporation of environmental factors into business models is important for helping businesses mitigate risks, improve operational efficiency, and increase competitiveness. Research by Eccles and Serafeim (2013) has shown that businesses with a sustainability focus achieve higher financial performance. Therefore, in this case, green finance is important for the transition to a sustainable economy, as it supports all sectors, not just those with energy-efficient production models. The role of government policies in the transition to a sustainable economy is important in building systems to support green finance and environmentally and economically sustainable development (Ye & Dela, 2023). In this case, green finance is important for the economic and social development of countries like China, which are transitioning to low-energy, low-pollution production models. It promotes energy-saving and environmental protection technologies that transform energy systems and economic systems (Schmidt et al., 2017). The green financial instruments include green bonds, village funds, equities, and sustainable banking loans, which provide innovative financial solutions with conducive terms to support sustainability initiatives (Ye & Dela, 2023). Therefore, it is instrumental in providing financing options to businesses, reducing operational risks, and driving economic transformation towards a green economy. The evaluation of green finance development is based on the adoption of green financial instruments, improvements in environmental protection, and the government's focus on green finance initiatives. Therefore, it promotes sustainable initiatives while driving economic transformation.

2.2.3 Institutional Theory

Institutional Theory, as brought to light by Fuenfschilling and Truffer (2014), has been recognized as a cornerstone in the science of organizations, with a focus on the significant impact of institutions, including rules and norms, on organizations. The theory suggests that institutions, including norms in society, culture, and rules, impact

organizations. Generally, Institutional Theory explains how higher-level rules, such as rules and norms, affect organizations, which are often considered to be common knowledge but are critical in understanding organizations. Generally, Institutional Theory suggests that organizations are not solely influenced by rationality and efficiency; they are significantly impacted by the pursuit of legitimacy in their environment.

In this case, legitimacy means the organization's actions are consistent with society's expectations. Organizations tend to align themselves with institutional pressure to achieve legitimacy, which is important for gaining acceptance and support from various stakeholders. This is because conformity to institutional pressure helps organizations deal with complexity and ambiguity through a socially accepted framework for decision-making. In relation to sustainable finance, the application of Institutional Theory would provide insights into how the financial sector's norms, values, and structures shape the promotion of sustainability practices (Tolbert & Zucker, 1983). For example, if the financial sector is embedded in an institutional environment that prioritizes short-term profit maximization over long-term sustainability goals, this would affect the promotion of sustainable finance instruments such as green bonds and sustainability-linked loans (Fuenfschilling & Truffer, 2014).

On the contrary, in environments with strong institutional support for sustainability, including supportive regulation for green finance or industry norms that favor sustainable investments, financial actors are more likely to consider environmental and social factors in their decision-making process (Tolbert & Zucker, 1983). In addition, Institutional Theory highlights the importance of institutional entrepreneurs, entities, or individuals who champion the adoption of new norms or practices within an existing institutional framework. Institutional entrepreneurs are at the heart of Institutional Theory, which emphasizes their pivotal role in driving institutional change towards sustainability by championing the adoption of new financial instruments or practices that align economic activities with environmental or social goals (Fuenfschilling & Truffer, 2014). Institutional entrepreneurs can drive change in institutional norms, making sustainability an integral part of the financial sector's operations or strategies. In essence, applying Institutional Theory to sustainable

finance enables different stakeholders to understand the intricacies of integrating environmental or social considerations into financial decision-making. By recognizing the importance of Institutional Theory, policymakers, regulators, or industry players can design initiatives that create a conducive environment for sustainable finance.

This might also include creating policies to encourage green investments, collaborating with financial institutions to develop sustainable products, and raising awareness of the importance of incorporating sustainability into financial strategies (Tolbert & Zucker, 1983). In conclusion, Institutional Theory has provided a robust framework for explaining the effects of institutions on financial behavior and decision-making, to promote sustainability in the financial industry (Fuenfschilling & Truffer, 2014). This has helped create an environment that supports sustainability, thereby promoting both sustainability and financial stability.

2.3 Policy Review

Nepal Rastra Bank has been instrumental in promoting the institutionalization of green banking principles by issuing policy guidelines and regulations. The Environmental and Social Risk Management (ESRM) Guidelines have been a landmark policy initiative in integrating environmental and social issues in the operations of the banking sector. The guidelines have encouraged banks to focus on financing sectors like renewable energy, agriculture, small and medium-scale industries, and environmentally friendly sectors, while discouraging environmentally hazardous investments (NRB, 2018). Such policy interventions have helped align the country's financial sector with the Sustainable Development Goals and climate change agreements, underscoring the banking sector's importance to the country's sustainable economic development.

While in place, the effectiveness of green banking policy in Nepal depends on the level of awareness among banking professionals and customers. Awareness of green banking directly affects stakeholders' awareness and willingness to adopt environmentally friendly banking behaviors. The literature indicates that customer and small business organizational awareness is a major factor affecting demand for

green financial products, thereby undermining the effectiveness of policy interventions (Shrestha, 2024). In terms of policy, awareness is critical to the effectiveness of policy interventions, as aware customers are likely to embrace digital, environmentally friendly financial products.

The importance of FinTech adoption has been highlighted in Nepal as an additional strategy to support green banking implementation. This is due to the rapid growth of mobile banking, digital wallets, online banking platforms, and QR-based payment systems, which have reduced the need to maintain physical banking infrastructure. Nepal Rastra Bank has supported this shift towards digital banking through payment system reforms, licensing of payment service providers, and the promotion of digital financial services to enhance efficiency and inclusion. This contributes to environmental sustainability by minimizing energy consumption and operational waste while improving bank efficiency.

From a profitability perspective, integrating FinTech services into green banking practices will enable Nepalese banking institutions to boost financial performance while achieving sustainability goals. This is because digitalization will help reduce transaction costs, improve risk monitoring, and increase access to markets that were previously underserved, thereby improving revenue growth potential. Empirical studies conducted in Nepal have shown that digital finance services positively influence financial inclusion and institutional performance, thereby supporting the argument that sustainability-focused innovations will boost financial profitability. Therefore, it can be argued that policymakers are increasingly viewing FinTech as a tool to balance sustainability with competitiveness.

Nevertheless, there are still some challenges to be addressed at the policy level to realize the full potential of green banking and FinTech. These challenges include differences in the use of FinTech services between urban and rural areas, limited integration of environmental performance metrics, and limitations among smaller financial institutions. Moreover, there are no metrics to assess green finance outcomes, making it hard to monitor policies (World Bank, 2020). This has to be addressed through coordinated policies that bring together financial sector regulations, financial technology, and environmental governance.

In conclusion, Nepal's policy environment shows a growing appreciation for the importance of green banking awareness and FinTech adoption in fostering profitable, sustainable banking practices. While the regulatory frameworks set by the NRB have already created a strong foundation in this regard, there is a need to further focus on these areas to help the financial sector in Nepal align with the concept of sustainability.

2.4 Empirical Review

Molla, Bhuiyan, Islam and Amin (2026) explored that this study intends to fill this gap by uncovering the factors conditioning the adoption of digital banks within the banking industry, embracing the digital bank led by the Technology Organization Environment (TOE) framework that links several conditioning variables in the banking industry, and how the digital bank is going to drastically change the landscape of Information and Communication Technology (ICT) resource utilization, maximize banking transactions, and bring a competitive advantage to banking performance. This study was conducted to gather the perspectives of employees of Bangladesh's banks on the topics mentioned. It also includes conducting pretests and pilot tests, addressing common method bias, using structural equation modelling, and performing analysis of variance tests to examine the research hypotheses related to the links among the factors being studied. Technological factors (RA and IRA, TBP), environmental factors (CP, LF, and GS), and organizational factors (TMS, TO, and EO) have 55.10% positive effects among respondents to the adaptation of digital banks in Bangladesh, which also leads to a 29.70% increase in banking or industry performance. The primary determinants of digital bank adoption are Relative Advantage (RA), technology-based products (TBP), Top Management Support (TMS), Competitive Pressure (CP), and legal frameworks (LF). There is a strong correlation between these criteria and the adoption of digital banking. ICT Resource Availability (IRA), Entrepreneurial Orientation (EO), Technology Orientation (TO), and Government Support (GS) have minimal impact on the adoption of digital banking. Banking companies need technological expertise in AI, machine learning, blockchain, and ICT to adopt digital banking for sustainable performance. The SMART economy, branchless services, and small loan facilities will improve the banking industry's performance. This study found that 41.0% of the impact

crystallizes in the final adoption of digital banks, influenced by technological, organizational, and environmental factors, which, in turn, affect bank industry performance (BIP).

Pria, Rubaie, and Prasad (2026) conducted a study that explores the evolving role of Financial Technology (FinTech) in supporting climate transition, focusing on its influence on sustainable development, environmental conservation, and socio-economic dynamics. It examines the interplay between FinTech innovations and climate goals, highlighting the opportunities and challenges in utilizing FinTech for sustainable urbanization, financial inclusion, and resource management. The chapter examines the influence of independent variables, including FinTech development, environmental transition, research and development (R&D), and policy frameworks, on achieving climate transition goals. Moreover, the results show substantial correlations between FinTech indicators and environmental factors, confirming FinTech's potential to support sustainable investments and reduce environmental risks. In addition, the chapter offers insights into aligning FinTech growth with sustainability and recommendations for integrating FinTech into sustainability strategies.

Irfan, Bhute, Yadav, and Dias (2026) explored the transformative role of artificial intelligence in advancing green finance by enhancing sustainable portfolio management practices. Through AI-driven data analysis, investors can prioritize environmentally responsible assets, such as green bonds and renewable energy funds, and align their portfolios with climate-positive goals. AI algorithms play a critical role in forecasting environmental risks, aligning investments with ESG (Environmental, Social, and Governance) standards, and fostering ethical finance trends. This study examines a range of AI tools that enhance decision-making in green finance, assessing their strengths and limitations and the societal benefits they can unlock. By aligning AI innovation with green finance objectives, this research advocates for a financial ecosystem that supports global sustainability and ethics.

Chakroborty, Banu, Alam, and Dobey (2026) noted that this study examines the influence of green finance and key financial indicators on the UN Sustainable Development Goals (SDGs) in Bangladesh's banking sector. Analyzing panel data

from 10 commercial banks (2011–2023), the study assesses the relationship between five SDGs and seven financial variables: green expenses, ROE, ROA, market value of shares, leverage ratio, dividend payout ratio, and R&D expenditure. Using OLS regression (EViews), the study finds that green expenses, ROE, and R&D positively and significantly affect SDGs, underscoring the role of sustainability and innovation. Conversely, ROA negatively impacts the SDGs, while leverage harms them, and dividend payouts hinder them. Market value shows minimal influence. The results emphasize the importance of green finance while cautioning against excessive asset-driven profitability and reliance on debt.

Nguyen, Truong, Tan, Dang, and Ngoc (2026) aimed to clarify the mediating roles of three factors—corporate governance, perceived efficiency, and financial performance—in the relationship between digital transformation and sustainable business practices among Joint Stock Commercial Banks (JSCBs) in Vietnam. Data were collected from 270 customers with experience using services at JSCBs and analyzed using Structural Equation Modeling (SEM) with the Partial Least Squares (PLS-SEM) technique. The results indicate that corporate governance and financial performance play statistically significant mediating roles in the relationship between digital transformation and sustainable business. By contrast, no evidence was found to support the mediating role of perceived efficiency in this relationship. These findings not only contribute to the existing body of knowledge on digital transformation and sustainability in the banking industry but also offer managerial implications for JSCBs in developing digitalization strategies to enhance competitiveness in an increasingly digital world.

The study conducted by Sharma, Vasishta, & Singla (2025) revealed that the rise of green banking (GB) as a significant driver of environmentally and socially responsible economic practices has gained substantial momentum in recent times. This paradigm shift in the banking sector has underscored the rise of green financial technology (FinTech), which is catalyzing the development of innovative technologies to promote environmentally friendly banking practices. The study examines the impact of green banking awareness (GBA) on green FinTech adoption (GFA), which in turn influences environmentally friendly investments, perceived profitability (PP), and sustainable banking (SB). The study employs a quantitative research design and partial least squares structural equation modeling (PLS-SEM).

The study uses online questionnaires administered to registered users of various FinTech products and services in the North Indian region, with 196 respondents. The study found a significant positive relationship between GBA and GFA. This indicated that greater green banking awareness would positively affect the adoption of green FinTech. Additionally, the research found a positive relationship between GFA and increased investments in ESG investments and the perceived profitability and sustainability of personal investment portfolios. This indicated GFA's potential to promote financial empowerment. This research contributes to the concept and practice of ESG investments at the individual level. It presents a new discourse and creates an Eco-Ed nexus framework. This is based on strategic insights to inform the implementation of transformative measures.

The study carried out by Rahman & Hossain (2025) focused on the idea that even though there is substantial research done to analyze the impact of green finance (GF) and green innovation (GI) on environmental performance (EP), little attention has been paid to the relationship between FinTech adoption (FA) and GF/GI, especially in the context of the COVID-19 pandemic. To achieve this objective, the study uses structural equation modeling (SEM) to analyze data collected from 302 banking professionals in Bangladesh, a developing country. The study's findings show that FinTech adoption has a significant impact on green finance, green innovation, and environmental performance. Additionally, the study found that green finance had a positive impact on green innovation and environmental performance. The study also found that green innovation partially mediates the relationships among FinTech adoption, green finance, and banks' environmental performance. As one of the earliest empirical studies to examine these interrelationships in the context of a global crisis, the paper contributes to the literature on technological innovation, green finance, and environmental sustainability in financial institutions in emerging markets. Furthermore, the paper's findings highlight the importance of FinTech adoption, green finance, and green innovation in improving banks' environmental performance and supporting sustainable economic development amid unprecedented crises.

Gupta, Modi, Das, and Sahu (2025) discussed the significant role financial technology has played in transforming banking. Financial technology has created new ways to achieve sustainable finance and sustainable development goals. Drawing on digital

banking services enabled by artificial intelligence, blockchain technology, big data analytics, and cybersecurity advancements, the review discusses how financial technology facilitates the development of sustainable finance. Sustainable digital banking offers solutions to traditional banking challenges, such as excessive carbon emissions, limited environmental finance product options, and low regulatory standards. The use of modern technical tools enables digital banking to deliver financial services to all, develop environmentally friendly investment services, and launch environmentally conscious loan services. The review discusses how financial technology facilitates the development of green financial products through digital payment services, green investment platforms, and central bank digital currencies. Both reviews discussed challenges to implementing sustainable digital banking. These challenges include regulatory, cybersecurity, and financial technology challenges. The research focuses on the current developments in green finance, including the integration of quantum computing and the implementation of IoT technologies in financial platforms. The paper draws on existing literature and practices in the field to provide an in-depth understanding of fintech innovations that promote sustainable financial systems. The research findings can serve as a guiding light for researchers, financial institutions, and policymakers in formulating strategies to promote sustainable digital banking. The paper also presents potential improvements to existing policies and future research directions to move towards sustainable financial institutions.

Ejaz, Bashir, Mahmood, and Wagan (2025) researched "Green Finance and FinTech Adoption to Examine the Mediating Role of 'Effects of Financial Literacy and Perceived Financial Benefits' in Sustainable Banking." Under this research title, the authors aim to investigate the effects of green finance on FinTech adoption in the context of sustainable banking, specifically regarding perceived financial benefits and financial literacy. With the global move towards environmental sustainability and the widespread adoption of financial technology, the integration of these two fields has become increasingly significant both theoretically and practically. Despite the growing importance of green finance and FinTech in the global financial landscape, there is limited research on the effects of cognitive and motivational factors on the adoption of green FinTech in emerging economies. This research aims to extend the Technology Acceptance Model (TAM) to examine the mediating effects of perceived

financial benefits and financial literacy on the relationship between green finance and FinTech adoption among banking professionals in Pakistan. The quantitative study used a cross-sectional survey design. Data was collected from 376 banking professionals in commercial and Islamic banks in Pakistan. To investigate the direct and indirect relationships among the constructs, Structural Equation Modeling (SEM) was used with SmartPLS 4. The findings show that green finance has a positive impact on FinTech adoption. Additionally, both perceived financial benefits and financial literacy are significant indirect effects, with perceived financial benefits playing a more important role in mediating this relationship. The findings suggest that users' cognitive abilities and motivational direction are important to consider to promote green FinTech adoption. The current study extends the technology acceptance model (TAM) by incorporating dual mediation into its framework.

In a study by Zaid, Khan, Al-Mekhlafi, Al Koliby, Saoula, Saeed, and Mohammad (2025), the focus was on the future of green finance, to explore its effectiveness through the lens of digital transformation and the sustainability of FinTech. The study was based on the notion that prior research on the relationship between green finance and its effectiveness in improving sustainable performance has been inconclusive; thus, it sought to examine the direct relationship between green finance and sustainable performance, as well as the moderating effects of FinTech adoption and digital transformation. The study surveyed 152 employees of Saudi Arabian banks and used hierarchical regression analysis with the PROCESS Macro. The study established a positive relationship between green finance and sustainable performance, with the adoption of FinTech significantly moderating this relationship. The three-way interaction analysis revealed that digital transformation serves as a higher-order moderator of FinTech adoption. These findings provide valuable insights into how digital capabilities shape the efficacy of green finance in emerging economies, underscoring the importance of integrating advanced digital technologies and infrastructure to leverage the sustainability benefits of green finance.

Yuan (2025) is a study grounded on the Resource-Based View (RBV) theory, which examines the influence of Fintech Adoption (FA) on Financial Performance (FINP) and Environmental Sustainability Performance (ESP) in Chinese banking institutions, with a focus on the mediating effects of Corporate Social Responsibility (CSR) and

Green Finance (GF). The study employed a combination of statistical analysis techniques, including Structural Equation Modeling (SEM) and Necessary Condition Analysis (NCA), to analyze data from 257 banking professionals. The study's empirical findings show that FA has a positive influence on FINP, ESP, CSR, and GF, with the most significant effect on GF initiative effectiveness, underscoring the importance of Fintech in the banking sector for achieving financial and environmental sustainability. The study, however, also showed that the direct effect of CSR on FINP is not significant, an area worthy of future research. The NCA analysis showed that FA and ESP are key factors in enhancing financial performance, underscoring the strategic importance of Fintech in sustaining a competitive advantage. This research contributes to the general knowledge on the overall impact of Fintech. It also provides policymakers and practitioners with useful guidance on integrating technological innovation with sustainable banking practices. The findings show that banks can use Fintech to improve financial and environmental sustainability, and policymakers should promote regulations to facilitate such integrations toward a sustainable future.

Tanchangya, Rahman, Siddiqi, Islam, Sarker, Naher, and Chowdhury (2025) conducted a study to assess green banking technology as a strategic tool to facilitate green finance for climate change mitigation in developing countries, particularly Bangladesh. This study aims to assess the adoption of green banking technology in Bangladesh using an extended UTAUT model. It identifies the key determinants of consumer intention: system quality, bank reputation, customer awareness, performance and effort expectations, and individual innovativeness. PLS-SEM is employed to analyze the results using 475 respondents. The results reveal that customer awareness, individual innovativeness, and bank reputation have a substantial influence on consumer intention, but performance expectancy and effort expectancy have no influence. Actual usage behavior is strongly predicted by behavioral intention. This study provides valuable insights for financial institutions and governments on how to encourage green banking to achieve climate change mitigation through technological advancements, consistent with Sustainable Development Goal 13 on climate change.

Joshi and Karmacharya (2024) designed a study to examine the impact of FinTech adoption on the sustainability performance of financial institutions in Nepal, with a

focus on the mediating roles of green finance and green innovation. The research design is descriptive and comparative, systematically outlining the characteristics of FinTech adoption among customers of financial institutions in the study area. The research has sought to establish the cause-and-effect relationships between the adoption of FinTech and sustainability performance. The study employed a structured questionnaire administered via an online survey platform to collect data from 180 respondents. The structured questionnaire consisted of a 5-point Likert scale. The study employed Confirmatory Factor Analysis (CFA) and Structural Equation Modeling (SEM) to validate the study's measurement model and to examine the research hypotheses. The study's findings have been outlined in a conceptual model, a graphical representation of the research outcomes. The respondents validated the study's findings: 80% confirmed the outcomes, while 20% did not—relationships between the constructs. Moreover, the study has confirmed that FinTech's impact on sustainability performance is positive, whereas its impact on green innovation is negative. The study has also shown, through the analysis, that the impact of FinTech on sustainability performance is not mediated by either green finance or green innovation. The study has added value to the existing literature by providing empirical evidence on the impact of FinTech on sustainability performance in Nepal. The study has provided policymakers and financial institutions with insights on how to integrate sustainability with technological advancement to create a positive impact. The study has filled a gap in the existing literature by providing insights into integrating FinTech with sustainable economic practices in Nepal to create a positive impact on sustainability, benefiting the environment, society, and the economy.

Serdarusic, Pancic, and Zavistic (2024) aim to investigate the influence of green finance on Fintech adoption and banking sustainability amid emerging trends in green banking practices and digital technologies. This quantitative research design collected data via a survey questionnaire administered to 304 participants, including bank customers and employees. The findings of the research respondents were analyzed using IBM SPSS for demographic analysis and SmartPLS for structural equation modeling. The results show that green finance has a significant influence on Fintech adoption and digital awareness. Additionally, digital awareness was found to influence Fintech adoption significantly. However, the direct influence of digital transformation on Fintech adoption was insignificant. This study also validated the

substantial impact of Fintech adoption on banking sustainability and revealed that digital awareness mediates the relationship between green finance and Fintech adoption. This study provides new insights into the relationship between sustainable finance practices and digital banking trends. It emphasizes the importance of digital awareness and green finance in banking practices. The findings of this study have important implications for banks in Croatia, including a strategic focus on digital awareness campaigns, the use of Fintech to deliver a better customer experience, and the promotion of collaboration to develop an enabling environment for Fintech.

In a study by Alsadoun & Alrobai (2024), the researchers examined the relationships among fintech adoption, green finance, green innovation, and sustainable performance in the United States financial sector. A cross-sectional study design was used, with data collected from a sample of 398 employees in entry-level and managerial positions. Convenience sampling was used to collect data, with a focus on obtaining a diverse sample of participants across different positions in the financial sector. The data analysis method used was Partial Least Squares Structural Equation Modeling. The study results indicate a positive relationship among fintech adoption, green finance, green innovation, and sustainable performance, highlighting the transformative power of fintech adoption for sustainability initiatives in the financial sector. The study indicates that fintech adoption has a positive effect on green finance and green innovation, while green finance and green innovation have a positive effect on sustainable performance. In addition, the adoption of fintech indirectly affects sustainable performance by shaping green finance and green innovation, underscoring the interconnected nature of these constructs in driving sustainability in organizations. The study's findings provide policymakers, financial organizations, businesses, innovators, and sustainability practitioners with important insights to leverage the potential of fintech and innovation to drive sustainability and enhance organizational performance in the financial industry.

Bhuiyan et al. (2024) conducted research to examine the impact of Fintech adoption in banks on environmental performance, focusing on employee efficiency and green finance initiatives such as green technology. The research design for this study used a survey to collect primary data. The research targeted a purposive sample of 258 banking employees from private banks. The study applied the partial least squares

method for overall data analysis. The study findings show that the adoption of Fintech in banks impacts employee efficiency, green finance, and green technology, thereby improving banks' environmental performance. The study provides insights into how Fintech adoption in banks can contribute to achieving broader ecological sustainability objectives in the banking sector by promoting the relationship among Fintech adoption, green finance, and technology development.

Zhou, Chau, Sibghatullah, Moslehpour, Tien, and Shukurullaevich (2024) examined the effects of green finance, which has been recognized as a form of financial support for green growth, significantly reducing greenhouse gas and air pollutant emissions. Green finance for agriculture, green buildings, and other green projects will enhance a country's economic growth. Advances in digital technology have revolutionized the financial industry. The emergence of fintech has the potential to improve the efficiency and competitiveness of financial systems and enhance financial inclusion. At the same time, as technology becomes more complex, there are potential risks. Thus, using a panel dataset of G7 and E7 countries, this research investigates the relationships among green finance, environmental benefits, total natural resource management, fintech, and economic sustainability. The panel dataset has employed rigorous estimation techniques, including quantile regression and the generalized method of moments (GMM) for long-run estimation. The cointegration between the variables is confirmed using the Johansen-Fisher panel test. In the long-run model, using the quantile regression method, the significance of environmental benefits and total natural resource management is evident. GMM is also used to confirm the study's findings, as it is more efficient at handling heteroscedasticity. The GMM findings confirm a negative relationship between GDP and fintech, green finance, and total natural resource management. At the same time, environmental benefits are positively related to GDP. The study provides more insight into the relationship between environmental factors and economic sustainability. Thus, it is of practical significance to the field of finance. Though the study is of significant value to the topic, some limitations are evident and can be addressed in future research.

Doghan and Chong (2023) discussed the research conducted on the major objective of this study, which is to "examine the role of green finance, green investment, and green innovation as mediators in the relationship between FinTech adoption and environmental sustainability." Moreover, the study aimed to "examine the moderating

impact of access to finance in this context, with a focus on manufacturing firms in Saudi Arabia." The study collected data from 300 employees working in small- and medium-sized manufacturing firms using convenience sampling. The research design used in this study is cross-sectional, and the analysis is conducted by using Partial Least Squares Structural Equation Modeling (PLS-SEM) in the quantitative research paradigm. The results showed that "the direct effect analysis revealed that FinTech adoption has a positive and significant impact on green finance, green investment, green innovation, and environmental sustainability." Furthermore, ecological sustainability was also found to be significantly increased through green finance, green investment, and green innovation. Another important aspect that emerged from the study is that access to finance positively and significantly moderates the relationship between green finance, green investment, and green innovation. The significance of this study lies in its conduct within the context of Saudi Arabian manufacturing firms. Thus, the study is a significant contribution to the existing body of knowledge. It also opens the door to further research in the area. The study provides policymakers and regulatory authorities with valuable insights into the importance of FinTech adoption for environmental sustainability.

In their study, Tian et al. (2023) sought to understand the impact of Fintech innovation and green transformational leadership (GTL) on corporate environmental performance (ENP) using the theory of ecological modernization (EMT) and the ability-motivation-opportunity (AMO) framework. Using a survey research design with 286 managers of manufacturing SMEs in Bangladesh as respondents, the study utilized a two-stage hybrid SEM-ANN method to analyze the collected data. The study revealed that Fintech innovation and GTL have significant positive impacts on green innovation. Furthermore, the study found that green innovation positively influences ENP as a mediator between Fintech innovation and ENP, as well as between GTL and ENP. This study extends EMT and AMO theories, offering practical implications for SME managers and policymakers to understand the significance of incorporating Fintech innovation and GTL to promote green innovation and improve environmental performance.

Mirza, Umar, Afzal, and Firdousi (2023), in their detailed research on the impact of technological innovations on business models, including financial services, reported

that the new technologies optimize different functional dimensions of banking firms. It is imperative to evaluate the potential of these new technologies to accelerate green finance while supporting bank profitability. The paper employs a comprehensive sample of European banks from 2011 to 2021. It uses a panel fixed-effects regression model to examine the relationship between the adoption of financial technology and bank profitability. The results indicate a positive relationship between investment in financial technology and green lending, driven by enhanced search, diligence, and monitoring efficiency enabled by new technologies. Similarly, the results indicate a direct influence of fintech investment on the risk-adjusted return on capital, associated with lower costs, an expanded product base, and lower economic capital. Findings also reveal the significant impact of firm size, human capital efficiency, and market concentration on determining bank profitability and green lending decisions. These results have important implications for the role of financial technology in green finance and sustainability-related goals.

Guang-Wen and Siddik (2023) explored the fact that despite the availability of substantial empirical evidence on the influence of green finance (GF) or green innovation (GI) on environmental performance (EP), only a few studies have attempted to examine the link between Fintech adoption (FA), GF, GI, and EP during the COVID-19 pandemic. Thus, by applying the structural equation modeling (SEM) approach to data from 302 banking staff in a developing economy (in this case, Bangladesh), this research empirically examines the associations among FA, GF, and EP, as well as the mediating role of GI. The empirical results indicated that FA has a significant impact on GF, GI, and EP, and that GF has a significant positive influence on GI and EP. Additionally, GI was observed to have a positive impact on EP and partially mediated the relationship between FA, GF, and EP in the banking sector. As one of the earliest studies to empirically investigate the relationships among these variables, these findings add to the existing scholarship on technological innovation, green finance, and environmental sustainability in the context of financial institutions in an emerging market during the pandemic. Moreover, the study highlights the importance of FA, GF, and GI in enhancing the EP of financial institutions and, ultimately, in promoting the country's sustainable economic development.

Naz, Asif, and Hameed (2023) conducted a study that found technology pervades every aspect of business operations, making it impossible for any organization to evade its presence. The customer's delight is now the organization's priority. Adopting the sustainable practice of using fintech to make customer interactions easier is now the game for every bank in Pakistan, and it requires serious attention. This study explored this gap and added the facilitation aspect of green banking practices to the loop. Without the Sustainable Development Goals (SDGs), no business sees its future growing. Adoption of fintech is one of them. The State Bank of Pakistan also promulgated green banking regulations in 2017, requiring banks to adopt green banking practices across their financial, operational, and daily operations. Our explanatory, quantitative study assessed the effect of fintech on sustainable green practices and the bank's green practice activities as a mediator through a questionnaire consisting of 31 questions filled by 302 banking professionals in Karachi, Pakistan, & found that the green banking activities mediated the relationship between the fintech adoption and sustainable banking practices but the direct effect of fintech is found to be insignificant as local environment. The employee's mindset is not ready for it. The study offers valuable insights and practical recommendations.

Zaidi and Shah (2023) examined that Fintech has the potential to revolutionize the financial sector in developing countries like Pakistan by providing access to financial services to the unbanked population. This article explores the role of fintech in improving Pakistan's financial situation. The main objective of this study is to identify how fintech can contribute to financial inclusion in Pakistan. The study is expected to contribute to the exploration of different fintech initiatives that may help bring the unbanked population into the formal financial sector. The study is expected to use a qualitative research approach, including a literature review to explore different fintech initiatives that may contribute to financial inclusion in Pakistan. The study has shown that fintech has the potential to improve Pakistan's financial position. The study has shown that fintech initiatives, such as mobile payments, digital banking, microfinance, and financial education, may help bring the unbanked population into the formal financial sector, especially in rural areas and among small business owners. The study has shown that this may help improve Pakistan's economy by providing access to financial services to those who lack them. However, it is important to address the challenges to finTech in Pakistan. Overall, this study's findings provide

insights into how finTech can contribute to economic growth in Pakistan and other developing countries.

Ravichandran and Rao (2023) emphasized that the high level of new technology and business model integration, along with the need to lower transaction costs and reach a broader audience as a result of the current wave of financial inclusion, have all contributed to the growth of the Fintech industry and economy. Regulators worldwide are also concerned about the potential impact of fintech on the financial sector. The high demand for upcoming fintech innovations, their applications and models, and the mitigation of any unfavourable repercussions that may result from such models. The authors examine the fintech sector in India and analyze its evolution and attendant repercussions. The situation in India has been quite encouraging, suggesting that the ecosystem is in a stage of maturation. Findings indicate that this may help India achieve financial inclusion, better integrate its economy towards the SDGs, enable faster growth through innovation, accelerate development, and provide a more robust and sustainable model. They offer their thoughts on how Fintech can help advance the economy after reviewing existing scholarship in this area, and they also present a possible interactive framework.

Eko, James, Inyang, Buwah, and Ntoh (2023) examined the green marketing practices and marketing performance of financial technology companies in Nigeria. It particularly focused on investigating the impact of green financial services, digital marketing, and electronic banking on the marketing performance of Fintech organizations. The study used a cross-sectional survey design. A structured questionnaire was used to collect primary data from 318 users of financial technology services in Calabar. In addition, descriptive statistics were used in the data analysis and interpretation, and the hypotheses developed were tested using multiple linear regression. The findings of this study revealed that green financial services, digital marketing, and electronic banking positively impacted the marketing performance of FinTech companies in Nigeria. Therefore, based on the findings of this study, we recommend that FinTech companies in Nigeria focus on sustainable practices by offering green financial services, promoting digital marketing, and providing electronic banking to customers to reduce environmental impact and promote a shift away from traditional banking systems. This is important because FinTech companies

in Nigeria can promote sustainable practices and green initiatives to enhance their marketing performance. Adopting such practices can play a critical role in improving marketing performance by appealing to green consumers. Given the limitations of this study, we suggest that further studies be conducted across more sectors of the Nigerian economy to demonstrate the potential of green marketing to enhance marketing performance in contemporary times.

Jaiwant and Kureethara (2023) examined the past few decades, during which a global technology revolution has occurred, and with this progression, a primary outcome has been environmental degradation. Even as the Earth faces ecological degradation and climate change, countries have begun to improve their financial systems. Curtailing financial expenses and optimizing the utilization of economic resources are now considered another solution to environmental challenges. The organizations' green and low-carbon transformation has now become the major business policy across the industries. Green finance is a financial approach and set of practices that reduce carbon footprints and promote more eco-friendly business models and practices. Business corporations, NGOs, researchers, academics, and society have recognized green finance as a catalyst for balancing our economy and nature. The world is designing a monetary system to create and develop eco-friendly systems that will reduce ozone layer depletion, control pollution, and increase green activities. This financial system would help protect the environment while growing organizational revenues. Green finance involves collecting economic resources to tackle climate change and environmental issues. It is a form of sustainable finance that focuses on ecological, social, and governance (ESG) issues and risks, aiming to increase long-term investments in sustainable economic activities and projects. This chapter will examine the role and significance of green finance in establishing a sustainable financial system to address environmental degradation issues. In this chapter, the term green finance will be defined broadly.

Udeagha and Muchapondwa (2023) found that the BRICS region has considered achieving environmental sustainability a top policy priority. Environmental distress is primarily caused by the region's continued reliance on fossil fuels to meet local energy needs. Additionally, the area has historically been a significant importer of fossil fuels, making it challenging to reduce its reliance on them substantially. As a

result, the BRICS nations' greenhouse gas (GHG) emission rates have steadily increased over time. Moreover, the region offers vast untapped renewable energy sources that can be used to generate power without harming the environment. In light of this, this paper examines the combined effects of green finance (GFN) and financial technology (fintech) on achieving the region's carbon neutrality goals from 1990 to 2020, while controlling for energy innovation, economic growth, and natural resource rents. The results for the BRICS economies, consistent with the EKC hypothesis, suggest that green finance, fintech, and energy innovation (GFN, fintech, and ENI) promote environmental sustainability. However, natural resources rent (NRR) and economic growth (GDP) degrade ecological quality. Additionally, bidirectional causality has been shown among CO₂ emissions, fintech, and NRR. However, GDP and ENI have been shown to exhibit unidirectional causality with CO₂ emissions. Based on the empirical findings, it is suggested that the BRICS countries should accelerate the development of green financial products, enhance the capacity of banks and financial institutions to offer green credit facilities, and conduct research on the application of GFN solutions.

Yan, Siddik, Yong, Dong Zheng, and Rahman (2022) examined the effect of FinTech adoption on the sustainability performance of banking institutions in an emerging economy, such as Bangladesh. Furthermore, this study examines the mediating roles of green finance and green innovation in the relationship between FinTech adoption and sustainability performance. To explore the relationships among the study variables, this study used data from 351 employees of banking institutions operating in Bangladesh during January to March 2021, employing a convenience sampling method. Furthermore, the study employed a two-stage structural equation modeling and artificial neural network (SEM-ANN) approach to analyze the data. The findings indicate that the adoption of FinTech has a significant impact on green finance, green innovation, and sustainability performance. Similarly, the results suggest that green finance and green innovation have a considerable positive influence on sustainability performance. Furthermore, the results reveal that green finance and green innovation fully mediate the relationship between FinTech adoption and the sustainability performance of banking institutions. Moreover, the present study makes a significant contribution to the existing literature on technological innovation, green finance, and sustainability performance, as it is the first study to examine both linear and non-

linear relationships among these variables using the SEM-ANN approach. As a result, the study emphasizes the significance of FinTech adoption, green finance, and innovation in achieving sustainable performance, while also highlighting the pressing need to integrate new technologies, green initiatives, and financing into banking strategies to support the country's sustainable economic development.

Zhang, Wang, Zhong, and Siddik (2022) conducted a study to examine the impact of green banking activities on green financing and on banks' environmental performance. It also identifies the mediating effect of green financing on the relationship between green banking activities and the environmental performance of private commercial banks (PCBs) in Bangladesh. Besides, this study also examines the significant challenges and benefits of green banking development in an emerging economy like Bangladesh. The convenience sampling technique was employed to collect primary data from PCB bankers in Bangladesh, yielding a final sample of 352. To assess the relationship among the study variables, the Structural Equation Modelling (SEM) approach was employed. The empirical results revealed that green banking activities have a significantly positive impact on banks' environmental performance and the sources of green financing, and that these sources of green funding have a significant influence on banks' environmental performance. Additionally, it was observed that green financing mediates the association between green banking activities and banks' environmental performance. Furthermore, the study identified customers' insufficient awareness of green banking, high investment costs, technical obstacles, a lack of capable staff to appraise green credits/loans, and difficulties and complexity in assessing green projects as significant challenges affecting the development of green banking in Bangladesh. Moreover, the study found that increasing banks' competitiveness, reducing long-term costs and expenses, providing online banking facilities, improving customer goodwill, and reducing carbon footprints are key benefits of green banking development, as they help achieve sustainable economic growth in the country. Therefore, the study's significant theoretical and managerial policy implications are further discussed, along with its limitations and future research directions.

Firmansyah, Masri, Anshari, and Besar (2022) found that the rise of financial technology (fintech) has been a substantial change in the economic landscape, driven

by technological advancements and the global financial crisis. This paper uses a systematic literature review (SLR) to examine recent research on fintech adoption/acceptance. The sources used are from the Scopus database (2019-2022). The last sources reviewed are sixteen journal articles published in various journals with different contexts and theories from different countries. Several inclusion criteria were used to filter the selected records. One crucial criterion is the journal's continuous presence in Scopus, which ensures the quality of the scholarly work it publishes. This criterion selection is expected to represent this paper's novelty. The study has shown different determinants based on the theories used by fintech researchers. Nevertheless, the most commonly used theories are the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT). In addition, trust, financial literacy, and safety are factors identified by previous researchers as important determinants of fintech adoption. Moreover, the study has shown that future research on fintech adoption should focus on developing a robust construct, given advances in fintech and changes in customer behavior.

Table 1: *Empirical Review of Related Studies*

Source (Author(s), Year)	Study Focus / Title	Objective of the Study	Key Findings
Molla et al., (2026)	Digital Bank Adoption and Banking Performance	To examine technological, organizational, and environmental determinants of digital bank adoption	Relative advantage, technology-based products, top management support, competitive pressure, and legal frameworks significantly enhance digital bank adoption and banking performance.
Pria et al., (2026)	FinTech and Climate Transition	To explore the role of FinTech in climate transition and sustainable development	FinTech development makes a significant contribution to sustainable investment, environmental protection, and climate transition.
Irfan et al., (2026)	AI and Green Finance	To examine the role of artificial intelligence in green finance and ESG investment	AI improves green portfolio management, environmental risk forecasting, and ethical finance.
Chakrobortty et al. (2026)	Green Finance and SDGs	To analyze the impact of green finance and financial indicators on SDG achievement	Green expenses and R&D positively affect SDGs, while excessive ROA and leverage negatively affect sustainability.
Nguyen et al. (2026)	Digital Transformation and Sustainability	To examine the mediating roles of governance and financial performance	Corporate governance and financial performance significantly mediate the relationship between digital transformation and sustainability.
Sharma et al., (2025)	Green Banking Awareness and Green FinTech Adoption	To assess the effect of green banking awareness on green FinTech adoption	Green banking awareness positively influences green FinTech adoption, ESG investment decisions, perceived profitability, and sustainable banking

Source (Author(s), Year)	Study Focus / Title	Objective of the Study	Key Findings
Rahman & Hossain, (2025)	FinTech, Green Finance, and Environmental Performance	To examine FinTech's effect on environmental performance	practices. Green innovation partially mediates the relationship between FinTech adoption and ecological performance.
Gupta et al., (2025)	FinTech-Driven Green Digital Banking	To review the role of FinTech in sustainable digital banking	FinTech supports ESG-oriented finance but faces regulatory and cybersecurity challenges.
Ejaz et al., (2025)	Green Finance and FinTech Adoption	To examine the mediating roles of financial literacy and perceived benefits	Financial literacy and perceived benefits significantly mediate the adoption of green finance and FinTech.
Zaid et al., (2025)	Digital Transformation and Green Finance	To examine the moderating roles of FinTech and digital transformation	FinTech adoption and digital transformation strengthen the relationship between green finance and sustainability.
Yuan (2025)	FinTech Adoption and Bank Performance	To examine CSR and green finance as mediators	Green finance mediates the impact of FinTech adoption on financial and environmental performance.
Tanchangya et al. (2025)	Adoption of Green Banking Technology	To identify factors influencing green banking adoption	Awareness, innovativeness, and bank reputation significantly drive adoption intention.
Serdarušić et al., (2024)	Green Finance and FinTech Adoption	To examine the role of digital awareness	Digital awareness mediates the adoption of green finance and FinTech, thereby enhancing the sustainability of banking.
Joshi & Karmacharya (2024)	FinTech Adoption in Nepal	To assess FinTech's impact on sustainability performance	FinTech improves sustainability performance, though mediation effects are weak in Nepal.
Alsadoun & Alrobai, (2024)	FinTech, Green Innovation, and Sustainability	To examine interrelationships among fintech, green finance, and	FinTech adoption positively influences sustainability through green finance and green

Source (Author(s), Year)	Study Focus / Title	Objective of the Study	Key Findings
		sustainability	innovation.
Bhuiyan et al., (2024)	FinTech and Environmental Performance	To examine the effect of FinTech on environmental initiatives	FinTech adoption enhances employee efficiency, green finance, and environmental performance.
Zhou et al., (2024)	Green Finance, FinTech, and Economic Sustainability	To examine environmental and economic impacts	Green finance and fintech can help reduce emissions, but may also hinder GDP growth.
Doghan & Chong (2023)	FinTech Adoption and Environmental Sustainability	To examine the mediating roles of green finance and innovation	Green finance, green investment, and green innovation significantly mediate sustainability outcomes.
Tian et al., (2023)	FinTech Innovation and Green Leadership	To analyze green innovation as a mediator	Green innovation mediates the effects of fintech and leadership on environmental performance.
Mirza et al., (2023)	FinTech Investment and Bank Profitability	To assess FinTech's effect on green lending	FinTech investment increases green lending and improves bank profitability.
Udeagha & Muchapondwa, (2023)	Green Finance, FinTech, and Carbon Neutrality	To examine the effect on CO ₂ emissions	Green finance and fintech can reduce emissions, while economic growth can increase them.
Guang-Wen & Siddik (2023)	FinTech, Green Finance, and Environmental Performance	To analyze the mediation of green innovation	Green innovation partially mediates the relationship between FinTech and environmental performance.
Naz et al., (2023)	FinTech and Sustainable Banking	To assess the mediation of green banking activities	Green banking activities mediate the relationship between FinTech adoption and sustainability.
Ravichandran & Rao (2023)	FinTech Ecosystem Development	To examine FinTech evolution and SDG alignment	FinTech supports financial inclusion and sustainable economic growth.
Eko et al., (2023)	Green Marketing in FinTech Firms	To examine green services and marketing	Green financial services and e-banking significantly enhance

Source (Author(s), Year)	Study Focus / Title	Objective of the Study	Key Findings
Zaidi & Shah (2023)	Role of FinTech in Promoting Financial Inclusion in Pakistan	performance To examine how FinTech contributes to financial inclusion in Pakistan	marketing performance. FinTech improves financial inclusion through mobile payments, digital banking, microfinance, and financial education; regulatory and digital literacy challenges remain.
Jaiwant & Kureethara, (2023)	Green Finance and Sustainable Financial Systems	To explore the role and significance of green finance in establishing a sustainable financial system	Green finance plays a crucial role in reducing carbon footprints, promoting ESG-based investments, and balancing economic growth and environmental protection.
Yan et al., (2022)	FinTech and Sustainability Performance	To examine the mediation of green finance and innovation	Green finance and green innovation fully mediate the relationship between FinTech and sustainability performance.
Zhang et al. (2022)	Green Banking Activities and Environmental Performance	To assess the mediation of green financing	Green financing mediates green banking activities and environmental performance.
Firmansyah et al., (2022)	FinTech Adoption Literature Review	To identify determinants of FinTech adoption	TAM and UTAUT are dominant theories explaining FinTech adoption.

2.5 Research Gap

The existing body of literature clearly demonstrates the independent contributions of green finance, green innovation, and FinTech adoption to environmental sustainability and organizational performance. Numerous studies have reported that green finance and green innovation play a significant role in enhancing environmental performance and supporting sustainable development in both financial institutions and manufacturing firms (Joshi & Karmacharya, 2024; Rahman et al., 2024). [In a similar](#)

vein, recent studies underscore the growing significance of FinTech adoption in improving operational efficiency, expanding financial inclusion, and advancing sustainable development (Gupta et al., 2025). Despite these pertinent findings, some gaps in essential research remain. To begin with, rather than exploring the cohesive, sequential relationship among FinTech adoption, green finance, and green innovation, existing research has largely treated these concepts as separate constructs. The joint facilitation of green finance and green innovation through FinTech adoption, as well as the processes by which both paths lead to environmental sustainability through mediation, have been the subject of only a limited number of empirical studies (Sharma et al., 2025). Consequently, little is known about how digital finance innovations translate into real environmental benefits.

Furthermore, an overall multi-mediator and comparative model is still missing, particularly in emerging and developing economies, despite several studies investigating the mediating effects of green finance or green innovation. Most available empirical data is based on single-country studies, such as those in China, Bangladesh, and Saudi Arabia (Hossain et al., 2024; Joshi & Karmacharya, 2024), which limits the overall applicability of the results across different institutional and regulatory settings. This represents a knowledge gap regarding the impact of contextual factors on FinTech's green initiatives.

In addition, empirical studies on the effects of moderating factors on the relationship between FinTech adoption and sustainability outcomes have been surprisingly absent. The moderating factors considered in this case include access to financing, regulatory support, and expertise in green banking. Although the existing literature acknowledges the potential importance of these characteristics, nothing is known about their moderating effects (Sharma et al., 2025). This creates a problem for assisting regulators and legislators in developing specific strategies to enhance the sustainability effects of FinTech adoption.

In addition, most of the literature has focused on sustainability at the organizational level, with little attention to individual or SME perspectives, especially in developing economies. Moreover, there is a need for more empirical data on the long-term implications of the COVID-19 pandemic for green finance and environmental sustainability, as it has driven digital transformation within financial institutions

(Rahman et al., 2024). Therefore, to create a more holistic and sensitive understanding of how FinTech could be used to support green finance, green innovation, and environmental sustainability, these gaps need to be filled.

The present study aims to fill a gap in empirical research on the contribution of green finance awareness and green FinTech adoption to sustainable and profitable banking practices in Nepal. While financial institutions in Nepal have started to implement green banking guidelines and sustainability-focused financial policies, there is a lack of research evidence to show that these practices are actually impacting customers' and investors' behavior positively or that they are being reflected in sustainable banking practices (Nepal Rastra Bank [NRB], 2022). Past research in developing countries has shown that green finance and FinTech can contribute positively to environmental sustainability and financial performance; however, these areas have not been studied in the Nepalese context (Muganyi et al., 2021). Further, most past research is descriptive and has not considered awareness, financial knowledge, and technology adoption within a single framework. Through these relationships, this study aims to address a critical research gap, provide contextualized research evidence for Nepal, and offer practical insights for policymakers and banking institutions seeking to advance sustainable, technology-driven financial systems.

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the research methodology employed in the study. It also presents an in-depth discussion of the procedures used to conduct the research. This chapter presents an overview of the research design and methodological orientation. It also discusses the rationale behind choosing a quantitative approach to investigate the relationships between the variables in the study. This chapter also presents information on the population and sample. It also identifies the target respondents and discusses their relevance to the research problem. The chapter also presents information on the sampling design, explaining how participants were selected. It also discusses how the sample size was adequate and appropriate.

Moreover, the chapter elaborates on the nature and sources of the data, with a focus on the difference between primary and secondary data and their respective contributions to the study. The research instruments used to collect the data, with a focus on the structured questionnaire, are discussed in detail, with reference to the scale of measurement and the use of existing research instruments to enhance reliability and validity. The analysis employed in the research is discussed, with reference to the statistical tools and techniques used to analyze the collected data and establish the suggested relationships between the variables. The research framework is discussed to create a conceptual illustration of the relationships between the independent, mediating, and dependent variables. Additionally, the study's variables are clearly discussed, with reference to their conceptual and operational definitions, thereby creating clarity throughout the research. The chapter therefore establishes a clear research methodology focused on transparency, rigor, and replicability.

3.1 Research Design

The research design of this study is a mixed-methods design that combines descriptive and causal-comparative research to examine awareness of green banking and FinTech adoption for profitable and sustainable business practices in Nepal. The descriptive research design will provide a clear picture of the current state of public awareness regarding green finance, green investment, and corporate social responsibility using simple statistical measures such as frequency distributions, percentages, and mean values. On the other hand, the causal-comparative design is used to examine causes and effects, particularly to determine how green financing and FinTech influence

sustainable business performance through efficiency, profitability, and environmental sustainability. This design is appropriate for Nepal because of the rapid development of financial technologies, which has been supported by various regulations to boost green growth, thereby ensuring that the results are not only representative of the current situation but are also applicable to future circumstances.

3.2 Population and Sample design

The population of this research would include individuals who are in any way related to the use of banking and financial services in Nepal, especially those with experience in green banking practices and financial services facilitated by FinTech. This would include customers, employees, and professionals working in commercial banks, development banks, and financial organizations that facilitate digital financial services such as mobile banking, internet banking, green financing, and so on. This group would be suitable for the research, as they have the experience and knowledge of green banking practices and the use of financial services in the Nepalese financial system.

In the absence of an exhaustive, easily accessible sampling frame that covers all users and practitioners of green banking and FinTech services in Nepal, this study uses a non-probability convenience sampling method. This sampling method is frequently used in FinTech and green finance studies, especially in developing countries, owing to time, cost, and accessibility constraints. This method of sampling involves selecting participants based on their availability to participate in the study, while considering their knowledge or experience with digital banking services.

To determine the sample size, this study uses Cochran's formula, which is appropriate for large populations. The formula is expressed as:

$$N = \frac{Z^2 PQ}{e^2}$$

Where;

N = Required sample size

Z = Z-value at the desired confidence level (1.96 for 95% confidence level)

P = Estimated proportion of the population possessing the attribute (0.5 is used for maximum variability)

$Q = 1 - P$

e = acceptable margin of error (0.05)

Substituting the values into the formula,

$$N = \frac{(1.96)^2 \times 0.5 \times 0.5}{(0.05)^2} = 384$$

Thus, a minimum of 384 respondents will be considered sufficient for the study. However, given the study's limitations, a slightly lower number of samples will be considered sufficient and acceptable from a statistical point of view, as recommended by earlier PLS-SEM studies, which considered a minimum of 400 observations sufficient for a reliable analysis.

Thus, the population considered for the study will provide a diversified view of green banking awareness and FinTech adoption, as well as methodological relevance in the Nepalese context.

3.3 Nature and sources of data and the instrument of data collection

This study employs a quantitative research design and collects primary data from respondents. The study collected data using a structured questionnaire designed by the researcher, informed by previous research, articles, and guidance from senior researchers. The study has used a Likert scale with five points to measure the perceptions of the respondents, which range from "strongly disagree" to "strongly agree," with scale points ranging from 1 to 5.

3.4 Method of analysis

Once the data were collected, they were systematically coded and entered into Microsoft Excel and SPSS for processing. The study used several statistical techniques to achieve its research goals. The study used descriptive statistics to organize the data, providing a general idea of the respondents' characteristics and their awareness of green banking and FinTech services.

Moreover, a correlation analysis was conducted to determine the nature and degree of association among the key variables under investigation. These variables included green banking awareness, FinTech adoption, green finance, and sustainable business performance. In addition, multivariate regression analysis was used to assess the combined impact of green banking awareness and FinTech adoption on sustainable and profitable business performance. In some cases, green finance was also included in the regression analysis. Using these analytical tools, an in-depth evaluation of the relationships among the variables was conducted, providing valuable insights into sustainable financial practices in Nepal.

Mediation Analysis: This study uses mediation analysis to assess whether Green FinTech Adoption acts as an underlying mediator, mediating the relationship between Green Banking Awareness and sustainable financial outcomes. Mediation analysis is used when the researcher wants to understand not only whether an independent variable affects one or more dependent variables, but also the process by which this effect occurs. Here, the researcher wants to understand if the effect of green banking awareness happens through green FinTech adoption before it affects investment decisions, expected returns, and profitability, as well as sustainable banking practices.

The mediation process adopted in this study was based on a regression analytical approach. The process involved a regression analysis to determine the relationships among the independent variable, the mediator, and the dependent variable. Firstly, the relationship was established to determine the impact of the independent variable on the mediator. Secondly, the direct relationship was established to determine the impact of the independent and mediator variables on the dependent variables. Thirdly, the mediator was added to the regression model to determine its impact on the dependent variables while controlling for the independent variable. Finally, an indirect relationship was established to determine whether mediation had occurred. A

reduction in the direct effect after controlling for the mediator indicated partial mediation, whereas a non-significant direct effect indicated complete mediation.

3.4.1 Reliability Test: Cronbach's Alpha (α)

A reliability test was conducted to assess the scales' reliability in measuring sustainable banking practices. Cronbach's alpha (α) was used to assess reliability, as it is strongly recommended for multi-item constructs (Cronbach, 1951). The results show that the sustainable banking construct, which incorporates environmental, social, and economic dimensions, demonstrates satisfactory reliability, as all Cronbach's alphas exceed 0.70, the required value for exploratory and explanatory research. This shows that the scales are reliable in capturing respondents' perceptions of sustainable banking practices. It is given by,

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum S_i^2}{S^2} \right)$$

Where,

- k = number of items in a scale
- S_i^2 = variance of each item.
- $\sum S_i^2$ = total variance of all items combined.
- If $\alpha \geq 0.70$ = acceptable reliability

3.4.2 Mean

The mean is a statistical measure of the average of a set of values. In this study, the mean value has been employed to measure the overall awareness level of green banking, FinTech, green finance, and sustainable business in Nepal's financial industry. By determining the mean response, this study has identified the overall trend in perceptions and awareness regarding green banking initiatives and the use of FinTech services to create sustainable, profitable business models. The higher the

mean value, the greater the awareness; conversely, the lower the mean value, the less awareness is observed. The mean is calculated using the following formula:

$$\text{Mean } (\bar{X}) = \frac{\sum X}{N}$$

Where,

X = Value of responses for each independent, mediating, or dependent variable.

N = Total number of responses.

This measure effectively summarizes the data and provides a clear understanding of the overall response pattern for the study variables.

3.4.3 Standard Deviation

Standard deviation, as a statistical measure, indicates the amount of variation or dispersion of the data values from the mean value. In this study, the concept of standard deviation has been employed as a measure of variation in respondents' perceptions of green banking awareness, FinTech adoption, green finance, and sustainable business performance in Nepal's financial sector. The higher the standard deviation, the greater the variation in respondents' perceptions of green banking awareness and FinTech adoption. On the contrary, a low standard deviation indicates more uniform perceptions among respondents.

The standard deviation is calculated using the following formula:

$$\text{Standard Deviation } (\sigma) = \sqrt{\frac{\sum (X - \bar{X})^2}{N}}$$

Where:

X = Value of responses for each independent, mediating, or dependent variable (green banking awareness, FinTech adoption, green finance, and sustainable business performance)

$\bar{X}$ = Mean value of responses for each independent, mediating, or dependent variable

N = Total number of responses

This measure helps assess the consistency of respondents' opinions and provides deeper insight into the degree of agreement or diversity in perceptions of the study variables.

3.4.4 Correlations Analysis

Correlation analysis is a statistical method that measures the relationship between two variables. In this study, a correlation analysis is used to examine the relationships among green banking awareness, FinTech adoption, green finance, and sustainable business performance in Nepal's financial sector. Correlation analysis will help determine whether increased green banking awareness and FinTech adoption are associated with business performance in Nepal. Additionally, the correlation will help determine whether the relationship is positive or negative. The correlation will provide valuable insights into the relationship between digital finance and green finance and their impact on business performance.

The correlation coefficient between two variables is calculated using the following formula:

$$r = \frac{n\sum XY - \sum X \sum Y}{\sqrt{n\sum X^2 - (\sum X)^2} \sqrt{n\sum Y^2 - (\sum Y)^2}}$$

Where:

n = Number of responses

X = Value of the independent variable

Y = Value of the dependent variable

This method enables the study to assess the degree of association among the key variables, supporting a clearer understanding of how green banking awareness and FinTech adoption relate to sustainability performance in Nepal.

3.4.5 Regression Analysis

Regression analysis is a statistical tool to investigate the impact of one or more independent variables on a dependent variable. In this study, multiple regression analysis was used to examine the impact of green banking awareness, FinTech adoption, green finance, and green innovation on sustainable, profitable business in Nepal's financial industry. This statistical tool helps to understand the impact of individual independent variables while keeping other independent variables constant. The regression model used in the study is expressed as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$$

Where,

- β = effect size (how strongly each factor influences Y).
- ε = error term.

3.5 Research Framework

The research framework of this study defines the systemic relationship between green banking awareness, green financial technology adoption, and sustainable banking. This framework includes an analysis of the relationship between green banking awareness and the adoption of green financial technology, thereby improving sustainability in the banking industry.

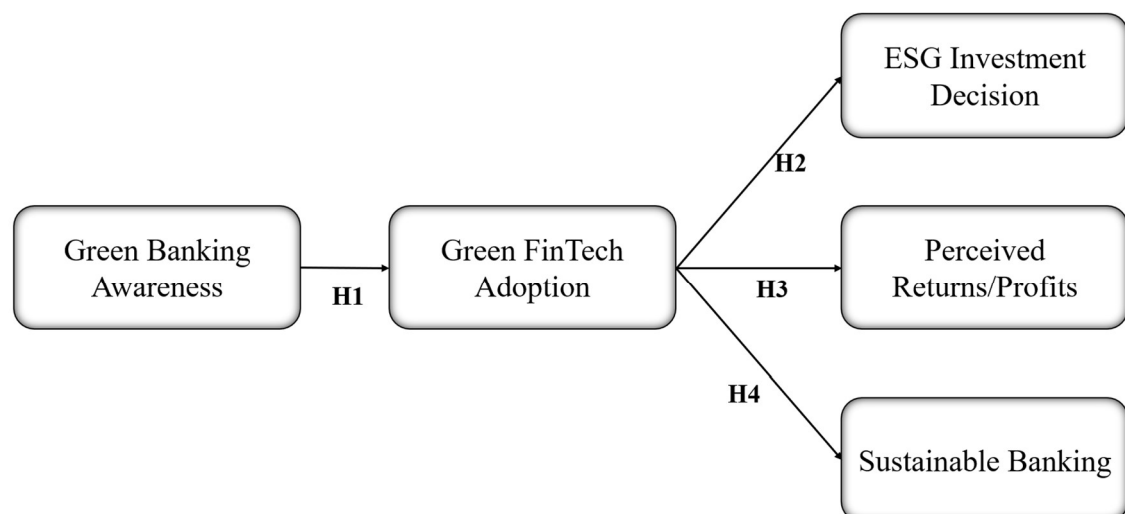
In this study, green banking awareness is treated as an independent variable. This concept includes the awareness of green banking principles, which focus on eco-friendly banking, paperless transactions, and investments in environmentally sustainable projects. Prior studies have shown that greater awareness of green banking principles is positively associated with green financial technology adoption, thereby improving sustainability in the banking industry. Green banking initiatives motivate both customers and financial institutions to support and engage in sustainable financial practices (Bansal et al., 2023; Bouteraa et al., 2023).

Green FinTech Adoption is proposed to mediate this model. It refers to the adoption of environmentally sustainable digital financial technologies. Such digital technologies include digital payment systems, online banking platforms, and green investment technologies. The existing literature suggests that FinTech acts as a

mediator in this scenario by transforming environmental awareness into an efficient financial solution (Muganyi et al., 2021; Yan et al., 2022). As a result, banks are more likely to adopt green FinTech to make operations more efficient and resource-saving.

The results of green FinTech adoption are measured using three dependent variables: ESG investment decisions, returns/profits, and sustainable banking. Green FinTech adoption is hypothesized to positively influence ESG investment decisions by increasing transparency, accessibility, and the evaluation of ESG-related information (Galeone et al., 2024; Lee et al., 2023). Moreover, green FinTech increases returns/profits by reducing operating costs and improving financial performance through digital efficiency (Mirza et al., 2023; Nenavath & Mishra, 2023). Finally, green FinTech has a positive influence on sustainable banking by promoting environmentally friendly activities, reducing carbon footprints, and aligning banking activities with long-term economic and social objectives (Guang-Wen & Siddik, 2022; Vergara & Agudo, 2021).

Overall, this research framework provides a logical approach to understanding how green banking awareness, facilitated by green FinTech, could be utilized to influence investment decisions based on ESG criteria, financial performance perceptions, and sustainable banking. This research framework is especially applicable to emerging economies such as Nepal, where financial innovations could play a vital role in moving forward sustainable development goals.



Source: Sharma, Vasishta, and Singla (2025).

Figure 1: Research framework of this study

Data Collection Instruments

Table 2: *Operational Definitions of Variables*

Conceptual Variable	Operational Definition	Measurement Indicators	References
Green Banking Awareness	The extent to which customers and stakeholders are informed about and understand environmentally friendly banking practices and initiatives that reduce carbon footprints and promote sustainability in the banking sector.	Awareness of green banking products; familiarity with environmental goals; understanding of environmental protection mechanisms; knowledge of environmental impact reduction; information received via media and bank communication	Bouteraa et al. (2023); Bansal et al. (2023)
Green FinTech Adoption	The degree to which financial institutions and customers implement and use financial technologies designed to promote environmental sustainability, reduce resource consumption, and support eco-friendly financial services.	Percentage of transactions conducted using green FinTech solutions. Adoption rate and frequency of use; Number of green FinTech products offered by banks; Customer satisfaction with green FinTech services	Muganyi et al. (2021); Aboalsamh et al. (2023); Yan et al. (2022)
ESG Investment	The practice of investing in companies and projects that meet environmental, social, and governance (ESG) criteria to achieve long-term financial returns and positive societal impact.	Integration of green FinTech into corporate strategies Importance of ESG compared to financial performance; willingness to recommend ESG investments; contribution to global sustainability; concerns about lower financial returns; perception of long-term risk reduction	Galeone et al. (2024); Lee et al. (2023); Asif et al. (2023)
Perceived Profitability	Stakeholders' perception of the financial benefits and returns derived from engaging in sustainable economic practices and green investments.	Strong financial returns; higher returns than alternatives; stability of profits over time; market-driven financial	Mirza et al. (2023). Lee et al. (2020); Nenavath

Conceptual Variable	Operational Definition	Measurement Indicators	References
Sustainable Banking	The implementation of banking practices and strategies that prioritize environmental sustainability through eco-friendly projects and the promotion of green finance initiatives.	performance; cost efficiency	& Mishra (2023)
		• Environmental integration in banking operations; support for community development; contribution to financial stability; influence on customer banking choice	Guang-Wen & Siddik (2022); Chueca Vergara & Ferruz Agudo (2021); Gupta (2015)

CHAPTER IV

RESULTS AND DISCUSSION

In this chapter, the study's findings are presented and analyzed to understand green banking awareness and FinTech adoption in promoting profitable, sustainable practices. The findings are presented in a clear, systematic manner to understand the patterns, relationships, and trends identified from the gathered data. Statistical tools are used to aid in understanding the findings holistically and to comprehend the impact of green banking awareness and FinTech adoption on promoting profitable and sustainable practices.

4.1 Results

After discussing and analyzing the findings, the chapter proceeds to the discussion section. In this section, the findings are compared with those of other studies on green financing and sustainable business practices. In this chapter, similarities and differences in findings are discussed in detail in comparison with other studies. This provides further insight into the contribution to knowledge relative to existing knowledge in the field. In this chapter, the findings are related to existing knowledge in the field to demonstrate their contribution to knowledge, in comparison with prior findings on green banking awareness and FinTech adoption in profitable and sustainable practices.

4.1.1 Demographic Profile of Respondents

This section presents a comprehensive analysis of the respondents' demographic characteristics. These include some of the most important characteristics, such as gender, age groups, and educational qualifications. Each of these has been closely scrutinized to provide a comprehensive picture of the respondents in this study.

The analysis of these characteristics will provide a better understanding of the sample represented in this study. This will be done to illustrate the various educational qualifications, professions, and experiences that are represented by this sample. It is important to understand these respondent profiles better, as this will be essential for interpreting the results of this study. This will ensure that results are interpreted within the context of diverse personal and professional characteristics.

Table 3
Demographic Profile of Respondents

Item	Frequency	Percent	Valid Percent	Cumulative Percent
Male	248	63.4	63.4	63.4
Female	143	36.6	36.6	100.0
Total	391	100.0	100.0	

Table 3 shows the distribution of respondents' genders involved in the research. From the total number of 391 respondents, 248 respondents (63.4%) were male, and 143 respondents (36.6%) were female.

The results show that the majority of the collected data consisted of male respondents. This is because they accounted for almost two-thirds of the respondents. The females accounted for slightly more than one-third of the respondents.

In conclusion, the data show that more respondents were male than female. This should be considered when interpreting the data collected. This is because it may affect the results.

Table 4
Demographic Profile of Respondents

Item	Frequency	Percent	Valid Percent	Cumulative Percent
Higher secondary education	117	29.9	29.9	29.9
Bachelor Degree	177	45.3	45.3	75.2
Masters	97	24.8	24.8	100.0
Total	391	100.0	100.0	

Table 4 presents the educational backgrounds of the 391 respondents in this study. From the data, it is evident that respondents have diverse educational qualifications, with most having higher educational qualifications.

The majority of respondents have a Bachelor's degree, comprising 177 people (45.3 percent of the total sample). This shows that nearly half of the sample have higher educational qualifications, which is the largest educational qualification among the sample.

A substantial number of people, 117 or 29.9 percent, have higher secondary educational qualifications. This shows that nearly a third of the sample has intermediate educational qualifications.

In addition, 97 people, or 24.8 percent, have a Master's degree. This shows that a substantial number of people have higher educational qualifications.

The cumulative percentages show that 75.2% of the respondents have at least a Bachelor's degree. This shows that most people are well educated. Overall, the findings suggest that most people have acquired some level of education, either at the undergraduate or postgraduate level. This could have helped them to be informed or knowledgeable.

Table 5

Demographic Profile of Respondents

Item	Frequency	Percent	Valid Percent	Cumulative Percent
Less than 20 Years	31	7.9	7.9	7.9
20-25 Years	116	29.7	29.7	37.6
25-30 Years	103	26.3	26.3	63.9
30-35 Years	59	15.1	15.1	79.0
35-40 Years	37	9.5	9.5	88.5
40 and Above	45	11.5	11.5	100.0
Total	391	100.0	100.0	

The table below shows the age distribution of the 391 participants in the study. From the results, it can be concluded that the participants are of diverse ages.

The largest number of participants falls within the 20-25-year age bracket. This group comprises 116 participants, which accounts for approximately 29.7% of the entire sample. This number is closely followed by the second-highest number of participants, who fall within the 25-30-year age bracket. This group comprises approximately 26.3% of the entire sample, consisting of 103 participants. This shows that the majority of the participants are young adults.

The number of respondents within the 30-35 years age bracket is 15.1%, representing 59 participants. The 40 years and above bracket comprises 11.5% of respondents, or 45 respondents. The 35-40-year-old bracket comprises 9.5% of respondents, or 37 respondents. The smallest bracket is the 20 years and below bracket, representing 7.9%, comprising 31 respondents.

The total percentages indicate that 79% of respondents are below 35 years of age, and 88.5% are below 40 years of age. This indicates that the study's respondents are mainly young people.

The study's findings indicate that the majority of respondents fall within the 20-30-year age bracket. This indicates that the study is mainly focused on young adults.

4.1.2 Reliability test

To assess the reliability of the collected data and maintain consistency across the study, this research used SPSS software for analysis. For this study, the reliability of the collected data is determined through Cronbach's Alpha. Cronbach's Alpha is a statistical method that measures the internal consistency of the items in each construct. The value of Cronbach's Alpha lies between 0 and 1. A value below 0.60 indicates low reliability, whereas values between 0.90 and 0.99 indicate excellent reliability. Generally, a value of more than 0.70 is considered acceptable for research purposes, as suggested by Nunnally (1978).

Table 6
Reliability Test

Variables	Cronbach's Alpha	No. of Items
Green Banking Awareness (GBA)	0.843	5
Environmental Social & Governance (ESG)	0.821	5
Perceived Profits (PP)	0.815	5
Sustainable Banking (SB)	0.811	5
Green FinTech Adoptions (GFA)	0.925	5
Overall	0.918	38

The table above shows the results of the reliability test conducted on the study variables. The values obtained from the reliability test, which include Cronbach's Alpha for GBA, ESG, PP, SB, and GFA, are 0.843, 0.821, 0.815, 0.811, and 0.925,

respectively. All the values obtained in this case exceed the minimum threshold of 0.70.

The overall reliability of all 38 items in this study is 0.918. The high value obtained in this case indicates the reliability of the scales used in this research.

4.1.3 Descriptive Statistics

In this research, descriptive statistical techniques have been employed to examine public awareness of green financing, green investment, and corporate social responsibility in Nepal. This technique would help summarize the data clearly and enable the audience to understand the level of familiarity people have with these essential concepts.

Using this technique, this research aims to provide a detailed insight into the level of awareness among the general public about environmentally and socially responsible financial practices. It would also provide insight into the level of public involvement in such practices. By examining these practices, the research aims to provide meaningful insights into how green practices are perceived in society.

4.1.4 Summary of Descriptive Statistics

In this section, descriptive statistics will be used to evaluate the overall nature of the key variables: Green FinTech Adoption (GFA), Green Banking Awareness (GBA), Environmental, Social, and Governance (ESG), Perceived Profitability (PP), and Sustainable Banking (SB). The analysis will use the mean and standard deviation for each variable to assess the nature of respondents' responses. Through this analysis, this study will provide a comprehensive overview of how respondents perceive green FinTech adoption, green banking awareness, environmental, social, and governance issues, profitability, and sustainable banking. This will be done using descriptive statistics to assess the nature of respondents' responses. This will provide an overview of how the respondents perceive green FinTech adoption, green banking awareness, environmental, social, and governance issues, profitability, and sustainable banking.

Table 7
Summary of Descriptive Statistics

Variables	Mean	Std. Deviation
GBA	3.4368	0.95072
GFA	3.7913	0.73252
ESG	3.4246	0.84539
PP	3.4875	0.85792
SB	3.5294	0.88783

The range of mean scores across all variables is 3.42-3.79. This suggests that the respondents agreed with the statements concerning green finance and sustainability practices. This is a moderately positive perception of all the constructs that were included in the study.

Among these variables, the Green FinTech Adoption variable has the highest mean of 3.7913. This suggests that respondents have relatively stronger acceptance and favorable attitudes toward green FinTech solutions. Furthermore, this variable has the lowest standard deviation value of 0.73252. This suggests that respondents have more consistent attitudes toward green FinTech adoption.

The Sustainable Banking variable has the second-highest mean value of 3.5294. The standard deviation of 0.88783 suggests a reasonable amount of variation in respondents' attitudes toward sustainable banking practices. However, the overall responses are relatively consistent.

The average Perceived Profitability (PP) score of 3.4875 indicates that respondents perceive green and sustainable initiatives as profitable. The standard deviation of 0.85792 indicates a balanced distribution of opinions. This suggests that although all the respondents agree on the financial viability of green and sustainable initiatives, they differ slightly in their perceptions about the financial returns. Green Banking Awareness (GBA) and ESG have average values of 3.4368 and 3.4246. This shows that they have moderate levels of awareness and positive perceptions about environmental, social, and governance practices. However, Green Banking Awareness (GBA) has the highest standard deviation value of 0.95072. This suggests that the levels of awareness of green banking concepts vary more among the respondents than those of the other variables. This may imply that although all the respondents are aware of green banking concepts, they differ slightly in their levels of awareness.

Overall, the results show that respondents have positive attitudes towards green financial practice in general and support the adoption of green FinTech in particular. The relatively moderate values of standard deviations across the scales suggest acceptable consistency in responses. Yet, the slightly higher values for green banking awareness suggest where awareness-building could be improved.

Table 8

Summary of Descriptive Statistics

Items	Mean	Std. Deviation
I have heard about green banking products offered by my bank.	3.30	1.236
I am familiar with the environmental goals of green banking.	3.42	1.176
I understand how green banking services help protect the environment.	3.51	1.123
I feel well-informed about how green banking reduces environmental impact.	3.50	1.150
I have received information about green banking through the media or bank communication.	3.45	1.156
N = 391		

Table 8 presents descriptive statistics on Green Banking Awareness based on data collected from 391 participants. The range of the mean item scores is 3.30 to 3.51.

The item "I have heard about green banking products offered by my bank" has a mean of 3.30 with a standard deviation of 1.236. The respondents have moderate levels of awareness. The item "I am familiar with the environmental goals of green banking" has a mean of 3.42 with a standard deviation of 1.176. The item "I understand how green banking services help protect the environment" has the highest mean (3.51) and the lowest standard deviation (1.123). The item "I feel well-informed about how green banking reduces environmental impact" has a high mean of 3.50 and a standard deviation of 1.150. The item "I have received information about green banking

through media or bank communication" has a mean of 3.45 with a standard deviation of 1.156.

All the items have moderate levels of awareness, which are relatively consistent with one another.

Table 9
Summary of Descriptive Statistics

Item	Mean	Std. Deviation
Green FinTech solutions are my preferred option for routine financial transactions.	3.9540	0.92150
The use of green FinTech services has increased over the past few years.	3.8926	2.65807
I trust green FinTech platforms to deliver reliable and secure services.	3.7033	1.05440
Management actively supports the adoption of green FinTech technologies.	3.7308	1.03751
Green FinTech services are easily accessible through my financial service provider.	3.6803	1.05616
N=391		

Table 9 presents the descriptive statistics for Green FinTech Adoption based on data collected from 391 participants. The mean score ranges from 3.68 to 3.95, indicating a positive perception of green FinTech adoption among the respondents.

The statement, "Green FinTech solutions are my preferred option for routine financial transactions," has the highest mean score of 3.9540, with a standard deviation of 0.92150. This indicates that the majority of the respondents prefer green FinTech for their financial transactions, and the data collected is consistent.

The statement "The use of green FinTech services has increased over the past few years" has a mean score of 3.8926 and a standard deviation of 2.65807. The statement "I trust green FinTech platforms to deliver reliable and secure services" has a mean of

3.7033 and a standard deviation of 1.05440. The statement “Management actively supports the adoption of green FinTech technologies” also has a high level of perceived support with a mean of 3.7308 and a standard deviation of 1.03751. Finally, the statement “Green FinTech services are easily accessible through my financial service provider” has a mean of 3.6803 and a standard deviation of 1.05616.

The results show that the respondents have a positive attitude towards green FinTech adoption. They also show moderate consistency in their attitudes.

Table 10

Summary of Descriptive Statistics

Item	Mean	Std. Deviation
ESG performance is as important as financial performance when evaluating investment opportunities.	3.37	1.080
I would recommend ESG-focused investment options to friends or colleagues.	3.54	1.073
ESG investments help address global sustainability challenges.	3.53	1.054
I am concerned that ESG investments may yield lower financial returns.	3.27	1.031
I believe ESG investments are less risky in the long term compared to 2n-ESG investments.	3.42	1.064
N=391		

Table 10 shows the descriptive statistics on ESG, with 391 participants responding to the statements. The mean values range from 3.27 to 3.54, indicating moderate agreement with the statements.

For the statement, "ESG performance is as important as financial performance when evaluating investment opportunities," the mean is 3.37 and the standard deviation is 1.08, indicating moderate agreement. The highest mean, 3.54, is observed for the statement "I would recommend ESG-focused investment options to friends or

colleagues," with a standard deviation of 1.073, indicating a positive attitude towards promoting ESG investments. The same is true with the statement "ESG investments help address global sustainability challenges," which has a mean of 3.53 and a standard deviation of 1.054, showing agreement with the statement.

The value of the item "I am concerned that ESG investments may yield lower financial returns" has a mean of 3.27 with a standard deviation of 1.031, implying a certain degree of concern regarding ESG investments. Lastly, the value of the item "I believe ESG investments are less risky in the long term compared to non-ESG investments" has a mean of 3.42 and a standard deviation of 1.064, suggesting moderate confidence in ESG investments.

Based on the above, it can be concluded that the respondents have moderately positive perceptions of ESG investments, acknowledging their importance and sustainability.

Table 11

Summary of Descriptive Statistics

Item	Mean	Std. Deviation
I believe this business/solution produces strong financial returns.	3.43	1.025
This option generates higher returns compared to alternatives.	3.48	1.012
Profits from this service appear stable over time.	3.51	1.015
The financial performance of this activity is driven by strong market demand.	3.51	1.059
Cost efficiency contributes significantly to profitability here.	3.49	1.062
N=391		

Table 11: Descriptive statistics of profitability based on the opinions of the 391 participants. The item means range from 3.43 to 3.51, indicating a moderate level of agreement with the statements regarding profitability.

The first statement, "I believe this business/solution produces strong financial returns," has a mean of 3.43 with a standard deviation of 1.025. The next statement,

“This option generates higher returns compared to alternatives,” has a mean of 3.48 with a standard deviation of 1.012. The third statement, “Profits from this service appear stable over time,” has a mean of 3.51 with a standard deviation of 1.015. The fourth statement, “The financial performance of this activity is driven by strong market demand,” has a mean of 3.51 with a standard deviation of 1.059. The fifth statement, “Cost efficiency contributes significantly to profitability here,” has a mean of 3.49 with a standard deviation of 1.062. The overall results show that respondents moderately agree with the statements on profitability; however, responses are consistent across all statements.

Table 12

Descriptive Statistics of Sustainable Banking

Item	Mean	Std. Deviation
This bank actively integrates environmental sustainability into its banking operations.	3.45	1.056
This bank's sustainability practices contribute to its financial stability.	3.57	1.033
This bank supports community development through its financial services.	3.54	1.024
The bank's sustainable banking practices influence my choice to use its services.	3.58	1.076
Sustainable banking contributes to the bank's long-term financial performance.	3.51	1.057
N=391		

Table 12 presents the descriptive statistics for sustainable banking based on responses from 391 participants. The mean scores range from 3.45 to 3.58.

The mean for the statement "This bank actively integrates environmental sustainability into its banking operations" is 3.45 with a standard deviation of 1.056. For the second statement, "This bank's sustainability practices contribute to its financial stability," the mean is 3.57 and the standard deviation is 1.033. For the third statement, "This bank supports community development through its financial

services," the mean is 3.54 and the standard deviation is 1.024. For the fourth statement, "The bank's sustainable banking practices influence my choice to use its services," the mean is 3.58 and the standard deviation is 1.076. For the fifth statement, "Sustainable banking contributes to the bank's long-term financial performance," the mean is 3.51 and the standard deviation is 1.057. Overall, the results indicate moderate agreement among respondents with the statements regarding sustainable banking.

4.1.5 Correlation Analysis

This segment of the study includes a correlation analysis that examines the relationships among Green FinTech Adoption (GFA), Green Banking Awareness (GBA), Environmental, Social, and Governance (ESG), Profitability (PP), and Sustainable Banking (SB). The main objective of the correlation analysis is to evaluate the relationships between the variables. Using correlation coefficients, it is possible to understand the extent of relationships among green financial practices, green banking awareness, environmental, social, and governance factors, and profitability in the context of sustainable banking. The findings of the correlation analysis provide a deeper understanding of the relationships between the variables.

Table 13
Correlations

		GBA	ESG	PR	SB	GFA
GBA	Pearson Correlation	1	.738**	.686**	.705**	.226**
	Sig. (2-tailed)		0.000	0.000	0.000	0.000
ESG	Pearson Correlation	.738**	1	.807**	.782**	.240**
	Sig. (2-tailed)	0.000		0.000	0.000	0.000
PR	Pearson Correlation	.686**	.807**	1	.832**	.319**
	Sig. (2-tailed)	0.000	0.000		0.000	0.000
SB	Pearson Correlation	.705**	.782**	.832**	1	.355**
	Sig. (2-tailed)	0.000	0.000	0.000		0.000
GFA	Pearson Correlation	.226**	.240**	.319**	.355**	1
	Sig. (2-tailed)	0.000	0.000	0.000	0.000	

****.** Correlation is significant at the 0.01 level (2-tailed). N = 391

The table above shows the results of the Pearson correlation analysis between GBA, ESG, PR, SB, and GFA. From the correlation analysis among these variables conducted with 391 respondents, it is evident that they are positively correlated. All these variables are positively correlated with each other.

Moving to GBA, the study reveals that GBA is positively correlated with ESG ($r = 0.738$), SB ($r = 0.705$), and PR ($r = 0.686$). The correlation coefficients indicate that as ESG, PR, and SB increase, GBA increases as well. The correlation between GBA is stronger for ESG than for SB and PR.

On the other hand, the correlation coefficient between GBA and GFA is .226. This shows a positive correlation, albeit weaker. The correlation coefficient indicates a weaker association between GFA and GBA. Analysis of the independent variables' correlation coefficients revealed that ESG, PR, and SB exhibit very high correlations. The highest correlation coefficient was found between PR and SB, i.e., .832. This was followed by the correlation coefficient between ESG and PR (.807) and, finally, the correlation coefficient between ESG and SB (.782). This shows that these variables are highly correlated. This shows that ESG, PR, and SB are highly correlated variables. These variables may be considered to be similar. However, it may be noted that, since the correlation coefficients among these variables exceed .80, there is a possibility of multicollinearity. Therefore, it would be appropriate to conduct further tests, such as a Variance Inflation Factor analysis.

About GFA, the results reveal moderate positive correlations with SB ($r = .355$) and PR ($r = .319$), while correlations with ESG ($r = .240$) and GBA ($r = .226$) are low but statistically significant. The implication here is that while GFA is positively related to all other variables, these relationships are not as strong as those obtaining among ESG, PR, SB, and GBA.

The results of the correlation analysis reveal that all variables are positively related, as they tend to move in the same direction. The strong correlations among ESG, PR, SB, and GBA are consistent with theory regarding these variables. The strong correlations

among some of the predictors underscore the importance of diagnostic evaluation in multivariate analysis.

4.1.6 Regression Analysis

In this section of the study, multivariate regression analysis has been used to evaluate the impact of Green FinTech Adoption (GFA), Green Banking Awareness (GBA), Environmental, Social, and Governance (ESG), and Profitability (PP) on Sustainable Banking (SB). This kind of analysis would help in evaluating the impact of independent variables on sustainable banking. The incorporation of multiple independent variables in this study would help evaluate the extent to which green financial adoption, green banking awareness, environmental, social, and governance factors, and profitability enhance sustainable banking.

Table 14

Model Summary of Regression Model

Model	R	R Square	Adjusted R-Square	Std. Error of the Estimate
1	.768 ^a	0.589	0.585	0.61311

The summary of the model, shown in Table 14 above, indicates the regression model's goodness of fit. The correlation coefficient, R, is 0.768. This indicates a strong positive correlation between the independent and dependent variables. The R-squared value is 0.589. This indicates that approximately 58.9% of the variation in the dependent variable has been explained by the model's predictors. The Adjusted R-Square value is 0.585. This is slightly lower than the R-squared value but higher, given the number of variables included in this model. The standard error of the estimate, which indicates the average difference between the observed and predicted values, is 0.61311. This indicates strong explanatory value and a good fit.

Table 15

ANOVA Table of the Regression Model

Model	Sum of Squares	df	Mean Square	F	Sig.
Regression	207.76	4	51.94	138.174	.000b

1	Residual	144.723	385	0.376
	Total	352.483	389	

Table 15 shows the result of the ANOVA test, which checks the significance of the regression model. The regression sum of squares is 207.76, and its degrees of freedom are 4. The regression sum of squares divided by its degrees of freedom gives us 51.94, which is the mean square. The sum of squares due to residuals is 144.723, and its degrees of freedom are 385. The mean square due to residuals is 0.376. The total sum of squares is 352.483, and its degrees of freedom are 389. The value of the F-statistic is 138.174, and its p-value is 0.000. Since the p-value is less than 0.05, the regression model is statistically significant, which means the independent variables have a significant impact on the dependent variable. The regression model has a good fit.

Table 16

Beta Coefficient of Regression Model

Variables	Unstandardized Coefficients		Standardized Coefficients		Sig.	Collinearity Statistics	
	B	Std. Error	Beta	t		Tolerance	VIF
1 (Constant)	0.883	0.168		5.249	0.000		
GBA	0.640	0.031	0.721	20.593	0.000	0.949	1.054
GFA	0.089	0.040	0.077	2.214	0.027	0.949	1.054

The regression analysis was conducted to evaluate the impact of Green Banking Awareness (GBA) and Green FinTech Adoption (GFA) on investment decisions in ESG. The results show that both factors have a positive, significant impact on ESG investment decisions. Green Banking Awareness shows a strong positive impact on investment decisions in ESG ($B = 0.640$, $\beta = 0.721$, $p < 0.001$), confirming that higher levels of green banking awareness would improve investment decisions in ESG. The significant standardized beta coefficient and high t-value ($t = 20.593$) show that the impact of Green Banking Awareness is the highest.

Green FinTech Adoption also has a positive, statistically significant impact on ESG investment decisions ($B = 0.089$, $\beta = 0.077$, $p = 0.027$). Though this impact is not as strong as that of Green Banking Awareness, it is important to note that Green FinTech Adoption is an important factor in ESG investment decisions.

Moreover, the values of Tolerance = 0.949 and VIF = 1.054 confirm that there is no multicollinearity among the independent variables in the model. This ensures the reliability of the results. The findings suggest that both Green Banking Awareness and Green FinTech Adoption are important factors in ESG investment decisions in the banking industry. However, Green Banking Awareness is more dominant in influencing sustainable investment behavior in the industry.

Table 17

Beta Coefficient of Regression Model

Variables	Unstandardized Coefficients		Standardized Coefficients		Sig.	Collinearity Statistics	
	B	Std. Error	Beta	t		Tolerance	VIF
1 (Constant)	0.713	0.180		3.953	0.000		
GBA	0.584	0.033	0.647	17.513	0.000	0.949	1.054
GFA	0.203	0.043	0.173	4.680	0.000	0.949	1.054

The regression analysis examined the impact of Green Banking Awareness (GBA) and Green FinTech Adoption (GFA) on Perceived Returns/Profits (PR). The findings show that both independent variables are positively and significantly related to the dependent variables. Green Banking Awareness was found to have a strong positive relationship with perceived returns/profits ($B = 0.584$, $\beta = 0.647$, $p < 0.001$). This implies that as Green Banking Awareness increases, so do perceived returns/profits. The high t-value ($t = 17.513$) further confirms that Green Banking Awareness is a strong predictor of perceived returns/profits.

Likewise, Green FinTech Adoption also shows a significant, positive relationship with perceived returns and profitability ($B = 0.203$, $\beta = 0.173$, $p < 0.001$). Although the impact is lower than that of Green Banking Awareness, the findings reveal that green FinTech adoption can significantly influence perceptions of financial performance.

Moreover, the results show that there is no multicollinearity among the independent variables in the model, as indicated by the collinearity statistics (Tolerance = 0.949; VIF = 1.054). The findings reveal that green banking awareness and green FinTech adoption both play significant roles in enhancing perceived profitability; however,

green banking awareness plays a more decisive role in influencing perceptions of financial performance in the banking sector.

Table 18
Beta Coefficient of Regression Model

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
	B	Std. Error	Beta			Tolerance	VIF
(Constant)	0.468	0.179		2.609	0.009		
GBA	0.615	0.033	0.659	18.57	0	0.949	1.054
1 GFA	0.25	0.043	0.206	5.804	0	0.949	1.054

Regression analysis was employed to examine the effects of Green Banking Awareness (GBA) and Green FinTech Adoption (GFA) on Sustainable Banking (SB). The results revealed that both GBA and GFA have positive, statistically significant relationships with sustainable banking practices.

Green Banking Awareness has a strong positive relationship with sustainable banking practices ($B = 0.615$, $\beta = 0.659$, $p < 0.001$). This reveals a strong positive relationship between green banking awareness and sustainable banking practices. The presence of a significant standardized beta coefficient and a high t-value ($t = 18.57$) reveals that green banking awareness has a strong predictive power for sustainable banking practices. This reveals that if financial institutions and their stakeholders are more aware of green banking initiatives, they are more likely to embrace environmentally responsible business practices.

Likewise, Green FinTech Adoption also shows a positive, statistically significant relationship with sustainable banking ($B = 0.250$, $\beta = 0.206$, $p < 0.001$). Although its relative importance is lower than Green Banking Awareness, the findings reveal that green FinTech adoption can make a significant contribution to strengthening sustainable banking practices. This reveals that technology integration plays a vital supportive role in enhancing sustainability in the banking sector.

In addition, the collinearity diagnostics (Tolerance = 0.949, VIF = 1.054) indicate no multicollinearity among the independent variables, suggesting that the regression equation is statistically reliable. The findings reveal that both green banking awareness and FinTech adoption make significant contributions to enhancing

sustainable banking practices, though green banking awareness plays a more decisive role.

4.1.7 Mediating variables

The mediation analysis examined whether Green FinTech Adoption acts as a mediator between Green Banking Awareness and the dependent variables. The role of a mediation analysis is to identify the mechanism by which an independent variable impacts a dependent variable through a mediator. In the current study, Green Banking Awareness has been used as the independent variable, Green FinTech Adoption as the mediator, and ESG investment decisions, returns/profitability, and sustainable banking practices as the dependent variables. The regression approach, along with bootstrapping, has been used to measure the impact. The indirect effect has been calculated using confidence intervals. The use of a regression approach, along with bootstrapping, provides a comprehensive understanding of the potential impact of awareness of green banking initiatives on financial and sustainability-related outcomes through the adoption of green FinTech.

Table 19

Mediating Role of Green FinTech Adoption Between Green Banking Awareness and ESG

Direct effect of X on Y

Effect	se	t	p	LLCI	ULCI
0.6403	0.311	20.5926	0	0.5792	0.7015

Indirect effect(s) of X on Y:

	Effect	BootSE	BootLLCI	BootULCI
GFA	0.0156	0.0174	-0.0025	0.0606

The analysis shows that the direct effect of X on Y is positive and statistically significant. The effect coefficient of 0.6403 shows that for every one-unit increase in X, Y will increase by approximately 0.64 units. The standard error of 0.0311 suggests that the result is precise with relatively low variability.

The t-value of 20.5926 is very high, indicating a strong effect of X on Y. The p-value of 0.0000, which is less than .001, shows that the result is statistically significant. Furthermore, the results show that the 95% confidence interval of the effect coefficient varies from 0.5792 to 0.7015. This range excludes zero, indicating that the effect of X on Y is statistically significant. The results show that X has a strong positive effect on Y.

In terms of the indirect effect of X on Y through GFA, the value is 0.0156. The bootstrap-based standard error is 0.0174. The bootstrap confidence interval range is from -0.0025 to 0.0606. Since the range includes 0, the indirect effect is not significant. This means that GFA does not mediate the relationship between X and Y.

Table 20

Mediating Role of Green FinTech Adoption Between Green Banking Awareness and Sustainable Banking

Direct effect of X on Y

Effect	se	t	p	LLCI	ULCI
0.6150	0.331	18.5698	0	0.5499	0.6802

Indirect effect(s) of X on Y:

	Effect	BootSE	BootLLCI	BootULCI
GFA	0.0434	0.0195	0.245	0.0944

The results show that the direct effect of X on Y is positive and statistically significant (Effect = .6150, SE = .0331, t = 18.5698, p = .0000). This implies that when X increases by one unit, Y also increases by .6150 units. The low standard error of .0331 indicates a low level of estimation error. Moreover, the t-statistic of 18.5698 indicates a strong relationship between X and Y. The 95% confidence interval for the direct effect lies between .5499 and .6802. Since zero is not included in this range, it can be inferred that the direct effect is statistically significant.

For the indirect effect of X on Y via GFA, the results indicate an effect size of 0.0434 with a bootstrap standard error of 0.0195. The 95% bootstrap confidence interval lies between .0245 and .0944. Since zero is not included in this range, it can be inferred that the indirect effect is statistically significant.

Table 21

Mediating Role of Green FinTech Adoption Between Green Banking Awareness and Perceived Returns/Profitability

Direct effect of X on Y

Effect	se	t	p	LLCI	ULCI
0.5835	0.333	17.5130	0	0.5180	0.6490

Indirect effect(s) of X on Y:

	Effect	BootSE	BootLLCI	BootULCI
GFA	0.0352	0.0197	0.0154	0.0868

The results indicate that the direct effect of X on Y is positive and statistically significant (Effect = .5835, SE = .0333, $t = 17.5130$, $p = .0000$). This suggests that for every one-unit change in X, Y changes by .5835 units. The standard error of .0333 represents a small estimation error. The t-value of 17.5130 also indicates a strong relationship between X and Y. The 95% confidence interval is between .5180 and .6490. Additionally, zero is not included in this interval; hence, the direct effect is statistically significant.

The indirect effect of X on Y through GFA is .0352 with a bootstrap standard error of .0197. The bootstrap confidence interval ranges from .0154 to .0868. Since zero is not included in the interval, the indirect effect is statistically significant. This indicates that GFA significantly mediated the relationship between X and Y.

4.1.8 Major Findings

A total of 248 respondents, representing 63.4 percent, are male, while 143 respondents, or 36.6 percent, are female. The age distribution reveals that 31 respondents, or 7.9 percent, are less than 20 years old; 116 respondents, or 29.7 percent, are between 20-25 years old; 103 respondents, or 26.3 percent, fall within the 25-30 years age bracket; 59 respondents, or 15.1 percent, are 30-35 years old; 37 respondents, or 9.5 percent fall within the 35-40; and 45 respondents; or 11.5 percent are 40 and above years old.

The majority of respondents hold a Higher Secondary Degree, with 117 individuals accounting for 29.9 percent. Those with a Bachelor's Degree, with 177 individuals accounting for 45.3 percent. Additionally, 97 respondents, or 24.8 percent, have a Master's Degree.

The mean for Green Banking Awareness is 3.4368, with a standard deviation of 0.95072. This shows that people have a generally positive attitude towards green banking awareness. However, there is some variability among people's perceptions. The mean for Green FinTech Adoption is 3.7913, with a standard deviation of 0.73252. This shows that people have a generally positive attitude towards green fintech adoption. This has led to responses that are not as diverse or dispersed. The mean for Environment, Social, and Governance (ESG) is 3.4246, with a standard deviation of 0.84539. This shows that people have a generally positive attitude towards ESG. However, there is some variability among people's perceptions regarding how much they value the impact or significance of ESG. The mean for Perceived Profitability is 3.4875, with a standard deviation of 0.85792. This shows that people have a generally positive attitude towards sustainable business practices. This has led to responses that are not as diverse or dispersed. However, there is some variability among people's perceptions regarding the impact.

Sustainable banking has a mean of 3.5294 and a standard deviation of 0.88783. However, this shows that respondents moderately agree with sustainable banking practices and their importance. The responses are relatively consistent with each other. In conclusion, the descriptive statistics show positive perceptions towards green banking awareness, the adoption of green fintech, ESG, and sustainable banking

performance. However, changes in responses indicate the level of agreement and the importance placed on these factors.

The results show a strong positive correlation between the two variables, with a value of .738. This shows a strong association between GBA and ESG: as ESG increases, GBA increases. Moreover, the value 0.000 is less than the level of significance 0.01. This shows that the results are not due to random chance; that is, the results are reliable. Therefore, this study rejects the null hypothesis. The results show a strong positive correlation between the two variables, with a value of .705. This shows a strong association between GBA and SB: as SB increases, GBA increases. The value of significance for this relationship is 0.000, which is less than 0.01 (two-tailed). This further confirms that the correlation between GBA and SB is statistically significant and not due to chance.

Therefore, the data indicate that the relationship between the variables is significant. Based on the results, a positive correlation exists between the two variables. The correlation coefficient for this relationship is $r = .226$. This indicates a weak positive relationship between GBA and GFA. This shows that as one variable increases, the other variable will increase. The significance value for this relationship is 0.000. This means the relationship between the variables is significant, since it is below the 0.01 level of significance. Therefore, the null hypothesis that there exists no relationship between the variables can be rejected.

Moreover, the results indicate a strong positive relationship between the two variables, with a correlation coefficient of $r = .686$. This indicates a strong correlation between GBA and PR, which means that as PR increases, GBA will also increase. The p-value of 0.000, which is less than the 0.01 level, indicates that the correlation between the two variables is statistically significant at the 1 percent level (two-tailed). This means that the correlation between the two variables is not due to chance; there is a real relationship between them. The model summary indicates an R value of 0.768, indicating a strong correlation among the predictors: green banking awareness, green fintech adoption, ESG, and sustainable banking.

The R-square value is 0.589, indicating that the joint effects of ESG, Perceived Profitability, and sustainable banking explain 58.9 percent of the variance in green

banking awareness performance. The adjusted R-squared value of 0.585 confirms the model's explanatory power.

The regression results show that Green Banking Awareness (GBA) has a substantial, statistically significant impact on all three dependent variables. When investment decisions are based on ESG considerations, Green Banking Awareness (GBA) was found to have a strong positive impact ($B = 0.640$, $\beta = 0.721$, $t = 20.593$, $p < 0.001$). This implies that a one-unit increase in green banking awareness results in a 0.640-unit increase in ESG investment decisions. In another model where investment decisions are based on Perceived Returns/Profitability (PR), Green Banking Awareness (GBA) was found to have a significant positive impact ($B = 0.584$, $\beta = 0.647$, $t = 17.513$, $p < 0.001$). In another model where investment decisions are based on Sustainable Banking (SB), Green Banking Awareness (GBA) was found to have a strong positive impact ($B = 0.615$, $\beta = 0.659$, $t = 18.570$, $p < 0.001$). This confirms that green banking awareness is the most dominant predictor in all models.

In Green FinTech Adoption (GFA), there are also positive, statistically significant results, albeit of lower magnitude. For ESG, GFA has a small influence ($B = 0.089$, $\beta = 0.077$, $t = 2.214$, $p = 0.027$). For PR, the influence is stronger ($B = 0.203$, $\beta = 0.173$, $t = 4.680$, $p < 0.001$). For SB, GFA has a notable role ($B = 0.250$, $\beta = 0.206$, $t = 5.804$, $p < 0.001$). The VIF value is 1.054, indicating no multicollinearity in the models.

Regarding mediation analysis, the direct effect of GBA on ESG remains significant (Effect = 0.6403, $p < 0.001$). In contrast, the indirect impact of GBA through GFA is 0.0156, with a confidence interval ranging from -0.0025 to 0.0606, including zero. However, for Sustainable Banking, the indirect impact is 0.0434, with a confidence interval of 0.0245-0.0944. Similarly, for Perceived Returns/Profitability, the indirect effect is 0.0352, with a confidence interval of 0.0154 to 0.0868.

The study's overall findings indicate that Green Banking Awareness has a strong direct impact on all dependent variables. However, Green FinTech Adoption acts as a mediator for the impact of Green Banking Awareness on Sustainable Banking and Perceived Returns/Profitability; however, it does not mediate the impact of GBA on ESG investment decisions.

4.2 Discussion

The primary objective of this research was to investigate the roles of green banking awareness (GBA) and green FinTech adoption (GFA) in enhancing sustainable financial practices and performance in the banking sector. The results strongly support the hypotheses presented in the research and show that both green banking awareness and green FinTech adoption indeed play an important role in enhancing sustainability-oriented performance. This aligns with recent literature on the importance of integrating green banking and green FinTech into modern financial systems (Alsadoun & Alrobai, 2024; Pria et al., 2026; Zaid et al., 2025). The research supports the notion that integrating green banking and green FinTech is becoming increasingly important for the development and performance of the banking sector.

First, the study revealed a positive and statistically significant relationship between green banking awareness and the adoption of green FinTech. The implication is that when financial institutions have more knowledge and understanding of green banking, they are likely to adopt green FinTech. The study supports Sharma et al. (2025), who found that awareness of green banking significantly influences the adoption of green FinTech, ESG investment, and sustainable banking. Tanchangya et al. (2025) supported this study by citing the importance of awareness and innovativeness in the adoption of green banking technology. In addition, Serdarušić et al. (2024) revealed the role of digital awareness in the adoption of green finance and FinTech, thereby confirming its influence on behavioral intention to adopt technology.

However, the moderate relationship suggests that while awareness may be a prerequisite for adopting FinTech, it may not be enough to ensure it becomes mainstream. Other factors, such as technological, structural, and institutional factors, are important influences on the diffusion of FinTech. For instance, Molla et al. (2026) showed that technological readiness, competitive pressure, and legal factors are significant factors that enhance the adoption of digital banking technology. Firmansyah et al. (2022) similarly showed that existing technology adoption models, such as TAM and UTAUT, explain the role of organizational factors in FinTech.

Secondly, the study's findings establish a positive and significant link between green FinTech adoption and investment decisions guided by ESG criteria. This study's

findings are consistent with prior research that has underscored FinTech's enabling role in fostering sustainable investment practices. For example, Chen et al. (2025) noted that FinTech facilitates sustainable investment by enhancing transparency, data analytics, and ESG monitoring. Moreover, Yan et al. (2022) established that green finance and green innovation fully mediate the link between FinTech and sustainability performance. Additionally, Rahman and Hossain (2025) noted that green innovation partially mediated the link between green FinTech adoption and ecological performance. Overall, this study's findings indicate that green FinTech plays an enabling role in fostering sustainable investment practices.

Third, this study validated a positive and significant relationship between green FinTech adoption and perceived returns and profitability (PR). This is consistent with Mirza et al. (2023), who proved that investment in FinTech has a positive impact on green lending and bank profitability. Moreover, Yuan (2025) demonstrated that green finance serves as a mediating factor between FinTech adoption and financial performance. Nguyen et al. (2026) demonstrated that financial performance mediates the relationship between digital transformation and sustainability. These studies proved that sustainable digital innovation has a positive impact on operational efficiency, cost reduction, and competitive advantage. However, Chakroborty et al. (2026) demonstrated that over-emphasizing financial metrics such as high returns on assets (ROA) and leverage can negatively impact sustainability.

Lastly, the study findings show that green FinTech adoption positively impacts sustainable banking (SB) practices. This is consistent with earlier research that supports the idea that FinTech boosts environmentally friendly banking practices. For instance, Naz et al. (2023) indicated that green banking practices mediate the relationship between FinTech adoption and sustainability. Similarly, Doghan & Chong (2023) and Tian et al. (2023) supported the idea that green finance and green innovation significantly mediate sustainability. Bhuiyan et al. (2024) argued that FinTech boosts environmental initiatives and employee efficiency. Further, Udeagha & Muchapondwa (2023) and Zhou et al. (2024) supported the idea that green finance and FinTech contribute to emission reductions and economic sustainability. Jaiwant & Kureethara (2023) emphasized that green finance is crucial for developing sustainable financial systems and reducing carbon footprints.

Overall, the results support the emerging consensus on the importance of green banking awareness and the adoption of green FinTech in driving financial sustainability. While awareness provides the cognitive and attitudinal foundation for green banking adoption, the integration of FinTech offers ESG-based decision support, economic profitability, and sustainable banking. The results support the emerging empirical evidence (2022-2026) on the interconnected nature of digitalization, green finance, and sustainability in modern banking systems (Alsadoun & Alrobai, 2024; Pria et al., 2026; Zaid et al., 2025).

CHAPTER V

SUMMARY AND CONCLUSION

5.1 Summary

The purpose of this study is to examine how increasing green banking awareness and green FinTech can contribute to improved sustainable practices and better financial performance in Nepal's banking industry. Nowadays, with everyone talking about climate change and going green, there is a clear need for a sustainable economic system. And green banking and green FinTech are certainly stepping up to the challenge. However, in Nepal, things are still evolving, as people continue to face challenges in this regard.

Our objectives are clear-cut. They are to determine how awareness of green banking influences the adoption of green FinTech, establish how this influences ESG-based investments, assess how it influences perceptions of returns and profitability, and determine how it influences sustainable banking.

We started with a solid literature review that covered the basics of green banking, green FinTech, ESG, profitability, and sustainability. This included a mix of concepts and theories, such as Green Finance Theory, Institutional Theory, Legitimacy Theory, and Sustainable Development, as well as a review of existing literature on digital finance, ESG, and related topics. We noticed a few apparent gaps, such as the lack of actual data from countries like Nepal on awareness, FinTech, and sustainability.

For the research, a descriptive-correlational study design was employed. The explanatory study measured people's perspectives on green awareness, FinTech, ESG, profitability, and sustainable banking. The correlational study, on the other hand, measured the correlation between green banking awareness (GBA), green FinTech adoption (GFA), ESG, perceived returns (PR), and sustainable banking (SB).

We zeroed in on our country, surveyed 391 banking and finance professionals through convenience sampling, and collected data through a well-designed, easy-to-

use questionnaire with a 5-point Likert scale ranging from "strongly disagree" to "strongly agree." The data analysis was conducted in Excel and SPSS, with descriptive statistics to provide an overview of the data, followed by Pearson correlations. The framework was based on GBA driving GFA, which, in turn, drove ESG, returns, and sustainability.

In addition, green banking awareness is positively and significantly related to green FinTech adoption. This means that the greater the green banking awareness, the greater the green FinTech adoption. Finally, it clicks that green banking awareness and FinTech innovation go hand in hand to promote sustainability in banking. Green FinTech is seen as a bridge between green banking awareness and positive outcomes in ESG investments, profitability perceptions, and sustainable banking. This is all happening right here in Nepal.

5.2 Conclusion

The primary aim of this research was to investigate how green banking awareness and green FinTech adoption influence sustainable financial practices and outcomes in the banking sector. The discussion section below interprets the findings in light of the research objectives and relates them to theoretical and practical implications.

This study began by establishing the impact of green banking awareness (GBA) on the adoption of green FinTech solutions (GFA). The study found a positive and statistically significant relationship between GBA and GFA. In essence, this means that the higher the awareness of green banking initiatives, the greater the likelihood of adopting green FinTech solutions. In other words, the better informed people and organizations are, the more likely they are to accept and adopt green FinTech solutions that advance sustainability. However, the strength of this relationship is relatively low, indicating that awareness alone is insufficient; technological, infrastructural, and literacy factors are important considerations.

The second objective was to investigate the impact of green FinTech adoption (GFA) on investment decisions based on ESG criteria (ESG). The results revealed a positive and significant relationship between GFA and ESG. Green FinTech platforms provide transparency in data, monitoring tools, and access to sustainable products,

thus encouraging ESG criteria in investment decisions. Although the results are statistically significant, they are not large enough to show that ESG decisions are driven solely by GFA; strategic considerations also influence them.

The third objective was to examine the impact of GFA on perceived returns and profitability (PR), which is applicable in banking. The analysis showed a positive and significant relationship between GFA and PR. Green FinTech is likely to improve banking operations efficiency, reduce costs through digitalization, and attract environmentally conscious stakeholders. The moderate relationship, therefore, indicates that various factors, including GFA, influence profitability in banking.

The fourth and final objective was to examine the impact of GFA on sustainable banking (SB) practices. The analysis showed a positive, significant relationship between GFA and SB. Green FinTech is likely to enable paperless banking, energy efficiency, transparent banking practices, and responsible lending.

These results are consistent with the research goals, which affirm GBA and GFA as important factors in sustainable banking finance. GFA has a positive impact on ESG investments, perceptions of profitability, and SB practices. The strength of relationships, however, indicates that sustainability is a multifaceted process that integrates awareness, technology, governance, and strategy for long-term performance.

Thus, this study contributes to the green finance literature by presenting empirical results from Nepal, demonstrating that improving GBA and GFA will enable banks to align with sustainability goals without compromising their viability.

5.3 Implications

5.3.1 Theoretical Implications

The present study contributes to theory in meaningful ways by deepening understanding of the relationships among green banking awareness, green FinTech adoption, investment decisions based on ESG factors, perceived returns and profitability, and sustainable banking practices. By employing a structured research

framework, the present study strengthens the theoretical relationship between environmental awareness, technological innovation, and sustainability.

The study's findings validate theoretical perspectives on the positive role of green banking awareness in enhancing green FinTech adoption, providing further evidence of the significance of knowledge and understanding in driving it. The study's findings also validate the theoretical assumption that green FinTech adoption enhances investment decisions informed by ESG considerations. The findings provide further evidence of the significance of green FinTech adoption in driving financial innovation, thereby enhancing transparency, accountability, and responsible investment practices. The study findings on the positive role of green FinTech adoption in enhancing perceived returns and profitability also validate the theory, which challenges the conventional belief that sustainability can negatively influence financial performance. The study's findings validate the complementary roles of sustainability and financial performance.

Furthermore, the strong association found between green FinTech adoption and sustainable banking practices provides further credence to the arguments forwarded in Green Finance Theory and Institutional Theory, which assert that technology and institutional channels are vital in promoting sustainable finance. By empirically substantiating these relationships in the Nepalese context, this study contributes to the theoretical discourse on sustainable finance in developing economies.

Overall, this study demonstrates the interconnected and multidimensional nature of awareness, technology adoption, ESG integration, profitability, and sustainable banking practices. This study contributes to the academic literature by examining the impact of digital green finance channels on organizational sustainability. It encourages further theorization to develop an integrated model that captures these complex relationships.

5.3.2 Practical Implications

The study's practical implications are substantial, especially for financial organizations, financial managers, financial regulators, and financial policymakers interested in fostering sustainability in the financial industry, particularly the banking

sector. The positive and significant relationships identified in the study demonstrate that green banking awareness and green FinTech adoption are not merely theoretical constructs but can be practically applied to boost sustainable banking performance.

The findings are important to banking practitioners in that they imply that raising green banking awareness among bank employees, customers, and stakeholders can motivate more people to use green FinTech services. Awareness campaigns, training schemes, and communication strategies can thus be very instrumental in promoting the digital sustainability transition.

The positive impact of green FinTech on ESG-based investment decisions shows that financial institutions should incorporate digital technologies to improve environmental and social risk assessment, monitoring, and reporting. The positive impact of green FinTech on ESG-based investment decisions shows that financial institutions should incorporate digital technologies to improve environmental and social risk assessment, monitoring, and reporting. The significant impact of green FinTech on perceived returns and profitability shows that sustainability and financial profitability are not zero-sum; thus, financial institutions adopting green FinTech can increase operational efficiency, reduce costs, and enhance their reputation and customer base, thereby boosting profitability.

In addition, the study also focuses on the importance of green FinTech in enhancing the adoption of sustainable banking practices. Financial organizations, in this case, need to focus on green FinTech innovations that help reduce environmental impact, conduct paperless transactions, and promote sustainable lending and investment practices. The study's findings, on the other hand, highlight the importance of establishing a supportive regulatory framework to enhance awareness of green banking and the adoption of green FinTech. This can be a crucial aspect in enhancing the adoption of a sustainable economic system in Nepal.

Thus, in conclusion, it may be said that the study has brought to the fore the importance of collective efforts by financial organizations and authorities to raise green awareness, promote technological uptake, and advance sustainable banking. All this, in turn, would contribute to the growth of sustainable banking in the country.

